
A Minimax Approach for Optimal Intervention Policy Learning with Two-Stage Outcomes

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Abstract

When designing interventions to promote desired actions, two-stage agent heterogeneity – encompassing both engagement with the intervention and completion of the desired action – creates significant challenges in identifying optimal intervention policies. While this two-dimensional heterogeneity creates distinct agent response types with varying marginal policy returns, existing literature typically falls short in full identification of all agent types, leading to inefficient intervention allocations. To address the challenge of learning optimal policies that account for two-stage outcomes, we propose a minimax approach within a counterfactual principal strata framework. A value function, accommodating varying policy returns across six potentially non-identifiable principal strata, is designed and partially identified to minimize the worst-case value loss relative to three benchmark policies: never-treat, always-treat, and oracle. We introduce three estimators for optimal policy learning: Principal Outcome Regression (P-OR), Principal Inverse Propensity Scoring (P-IPS), and Principal Doubly Robust (P-DR), providing theoretical guarantees for their unbiasedness, robustness, and regret upper bounds. Extensive numerical experiments demonstrate the effectiveness and superiority of the proposed approach.

1. Introduction

When policymakers implement interventions to promote desired actions (e.g., educational programs to improve student outcomes, preventive healthcare reminders to boost

treatment adherence, or promotional incentives to increase customer purchases), a fundamental challenge arises in intervention design due to agent-level heterogeneity in two critical responses: (i) engagement with the intervention itself (students enrolling in small-group courses, patients attending check-up appointments, or customers collecting discount offers), and (ii) completion of the desired action (improved long-term student outcomes, medication compliance, or actual product purchases). Crucially, these responses jointly determine policy effectiveness: while some agents only act when receiving (and engaging with) the intervention, others may act regardless of intervention, making intervention allocation wasteful. Moreover, certain individuals might engage with the intervention yet fail to act, or conversely, take the desired action without formal engagement. This two-dimensional heterogeneity creates distinct agent response types, each generating different marginal policy returns (Wu et al., 2024). Beyond the aforementioned domains – including education, healthcare, and e-commerce – this framework also applies to recommendation systems, financial inclusion programs, and many other settings where intervention efficacy depends on sequential agent responses (Ma et al., 2018; Dwivedi et al., 2025).

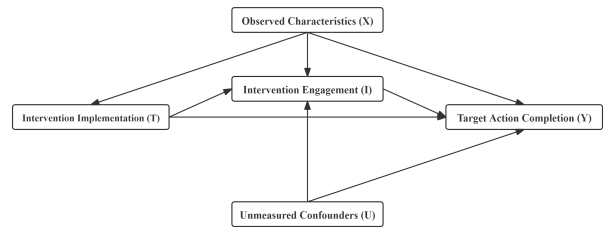


Figure 1. The causal diagram of Intervention Implementation $\rightarrow$ Intervention Engagement $\rightarrow$ Target Action Completion.

As formalized in Figure 1, we frame this problem as a causal classification task¹, where optimal intervention targeting requires accurate prediction of both stages of agent response. The analysis confronts an intrinsic limitation: outcomes

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under intervention and non-intervention conditions cannot be simultaneously observed for the same individual. Consequently, the analysis relies on counterfactual definitions of how individuals would have responded under alternative, unobserved intervention conditions. These are typically referred to as counterfactual individual responses (Li & Pearl, 2019). While existing literature has focused primarily on counterfactual individual responses in single-outcome scenarios (Ben-Michael et al., 2024; Li & Pearl, 2024; Li et al., 2023a; Wu et al., 2026), we aim to learn joint counterfactual responses that incorporate both the first-stage variable I and the second-stage variable Y , which will be rigorously formalized in Section 3. Based on the joint counterfactual outcomes of (I, Y) , individuals can be classified into distinct principal strata, generating varying policy returns. Policymakers seek to design optimal policies that prioritize allocating interventions to individuals expected to yield higher returns.

A central challenge stems from the non-identifiability of principal strata scores (i.e., the probability of an individual with certain characteristics belonging to specific counterfactual response stratum), which poses significant challenges for optimal policy formulation. Prior work (Li et al., 2023b) addresses the issue by restricting the analysis to specific application domains, such as e-commerce, and imposing exclusion assumptions that posit the absence of certain principal strata. However, such assumptions often fail in practice (Li et al., 2023b), limiting their generalizability. Without these assumptions, principal strata scores can only be partially identified from the data, leading to interval estimates with upper and lower bounds (Ding & Lu, 2016).

To address challenges of learning an optimal policy with two-stage individual responses and non-identifiable principal strata scores, we propose a minimax approach within a counterfactual principal strata framework. We define a value function to measure policy returns and develop a minimax approach that partially identifies this function, minimizing the worst-case value loss relative to three benchmark policies: never-treat, always-treat, and oracle. We introduce three estimators – Principal Outcome Regression (P-OR), Principal Inverse Propensity Scoring (P-IPS), and Principal Doubly Robust (P-DR) – and provide theoretical guarantees for their unbiasedness, robustness, and finite-sample regret upper bounds. The contributions of this paper are summarized as follows:

- We propose a novel minimax policy learning approach within a counterfactual principal strata framework to address the challenges of learning an optimal policy in non-identifiable scenarios.
- Our approach operates under relatively relaxed assumptions (i.e., without the exclusion assumption and any restriction on principal strata scores), ensuring its applica-

bility across diverse scenarios. Moreover, the proposed framework can be easily extended for further relaxations.

- We provide comprehensive theoretical analyses of the proposed method, demonstrating the unbiasedness and robustness of the estimators, and establishing an upper bound for the finite-sample regret of the learning strategy.
- Extensive experiments based on real-world data demonstrate the practical utility and effectiveness of the proposed method.

2. Related Work

Evolving beyond simply estimating heterogeneous intervention effects, a substantial portion of recent work has focused on estimating policy functions to determine the optimal intervention assignment based on pre-intervention features (Athey & Wager, 2021; Li et al., 2024; Kallus & Zhou, 2021; Zhang & Bareinboim, 2018). The topic of policy learning from observational data has been explored across both statistics (Luedtke & Van Der Laan, 2016; Qian & Murphy, 2011; Zhang et al., 2012; Athey & Imbens, 2017; Kitagawa & Tetenov, 2018) and machine learning (Swaminathan & Joachims, 2015; Dudík et al., 2011; Athey & Imbens, 2019; Xu et al., 2020).

Many causality-based methods have been developed, including outcome regression (OR) (Imbens & Rubin, 2015), inverse propensity scoring (IPS) (Austin & Stuart, 2015), and doubly robust (DR) approaches (Funk et al., 2011; Glynn & Quinn, 2010). A promising direction involves applying machine learning techniques to causal inference problems. For instance, Imai & Ratkovic (2014) use LASSO to identify predictive interactions between intervention and covariates, while Wager & Athey (2018) employ random forests to estimate heterogeneous intervention effects. For the two-stage outcomes scenario, Imbens et al. (2024) study the identification and estimation of long-term intervention effects by combining short-term experimental data and long-term observational data. Additionally, Yang et al. (2024) propose a principled policy learning approach for balancing multiple long-term and short-term returns.

Some recent studies have considered the joint set of potential outcomes. Li & Pearl (2024) study the unit selection problem. Ben-Michael et al. (2024) consider optimal policy learning algorithm with asymmetric counterfactual utility functions that incorporate the joint set of potential outcomes. Li et al. (2023a) propose a policy learning approach that satisfies the counterfactual no-harm criterion. Freeman et al. (2022) propose a algorithm to learn an optimal dynamic intervention regime with the principal strata scores.

In the context of optimal intervention policy learning, existing literature typically falls short in full identification of all principal strata and often assumes a uniform policy re-

Table 1. The counterfactual principal strata framework.

STRATA	DESCRIPTION	I(0)	I(1)	Y(0)	Y(1)	RETURN	LABEL
Y_{0000}	NO CLAIM - NEVER BUYER	0	0	0	0	r_1	l_1
Y_{0001}	NO CLAIM - INCENTIVE BUYER	0	0	0	1	r_2	l_2
Y_{0011}	NO CLAIM - ALWAYS BUYER	0	0	1	1	r_3	l_3
Y_{0100}	INCENTIVE CLAIM - NEVER BUYER	0	1	0	0	r_4	l_4
Y_{0101}	INCENTIVE CLAIM - INCENTIVE BUYER	0	1	0	1	r_5	l_5
Y_{0111}	INCENTIVE CLAIM - ALWAYS BUYER	0	1	1	1	r_6	l_6

turn, leading to inefficient intervention allocation. In this paper, we extend previous research and introduce a novel framework that employs a minimax approach within a counterfactual principal strata framework, aiming to address the challenges associated with optimal intervention policy learning.

3. Preliminaries

3.1. Motivating Example

As noted in Section 1, our proposed framework demonstrates broad applicability in many fields. Here we illustrate the concept with a concrete case example. Consider scenarios where e-commerce platforms issue incentives (e.g., coupons, cash rewards, or discounts) to users, which constitute interventions that typically incur non-negligible implementation costs. Here, T represents whether the incentive is offered, I indicates whether the user claims the incentive, and Y denotes whether the user makes a purchase. Based on the joint counterfactual outcomes of (I, Y) , individuals can be classified into distinct principal strata, generating varying policy returns. For example, consider three types of users:

- Type A: Purchases only after claiming the incentive.
- Type B: Purchases upon incentive exposure without claiming (e.g., price-insensitive users deterred by claim effort and incentive functions as advertisement).
- Type C: Purchases regardless of incentive but claims when available.

When issuing incentives:

- Type A generates modest revenue.
- Type B delivers greater values, offering higher profit margins (no discount cost) and demonstrating long-term revenue potential due to strong purchasing power.
- Type C creates negative returns: incentives incur costs without driving additional purchases, and withholding them later may lead to attrition (e.g., discontinuing purchases if incentives cease).

Thus, the platform’s optimal policy prioritizes Type B, then Type A, while excluding Type C.

3.2. Notations and Assumptions

Suppose that we have a simple random sample of n units from a super population $\mathcal{P}$ where each unit $i = 1, \dots, n$ has a set of characteristics $X_i \in \mathcal{X}$ and we consider a binary intervention assignment decision $T_i \in \{0, 1\}$ for unit i . Specifically, $T_i = 1$ if the intervention is administered, and $T_i = 0$ otherwise. We formally define the first-stage variable (intervention engagement) I and the second-stage variable (target action completion) Y as binary indicators of sequential outcomes for each unit i , i.e. $I_i, Y_i \in \{0, 1\}$.

To examine the counterfactual actions of units in response to interventions and their subsequent behaviors of interest, we adopt a potential outcome framework (Imbens & Rubin, 2015). let $(I_i(1), Y_i(1))$ and $(I_i(0), Y_i(0))$ be the potential outcomes for the unit i with and without intervention, respectively. The observed values of I_i and Y_i are deterministic functions of both the intervention assignment and the potential outcomes, which is known as the consistency assumption in the causal literature (Peters et al., 2017; VanderWeele, 2009). Specifically, $I_i = (1 - T_i)I_i(0) + T_iI_i(1)$ and $Y_i = (1 - T_i)Y_i(0) + T_iY_i(1)$. We assume that the stable unit treatment value assumption (SUTVA) (Ding, 2024) holds, i.e., there should not be alternative forms of the intervention and interference between units. In this study, we also assume the following assumptions:

Assumption 3.1 (Strong Ignorability and Strict Overlap). $\{I_i(0), I_i(1), Y_i(0), Y_i(1)\} \perp\!\!\!\perp T_i | X_i$ and there exists an $\eta > 0$ such that $\eta < e(x) < 1 - \eta$ for all $x \in \mathcal{X}$, where $e(x) \equiv P(T_i = 1 | X_i = x)$ represents the propensity score.

Assumption 3.2 (Monotonicity and Constant Control Potential Outcome). $Y_i(1) \geq Y_i(0)$ and $I_i(0) = 0$ for all unit i .

Assumption 3.1 is standard in observational studies (VanderWeele, 2008) and Assumption 3.2 is lenient and widely applicable (Li et al., 2023b; Jiang et al., 2016). It is noted that our proposed framework and methodology remain valid without Assumption 3.2, though the theoretical analysis would become substantially more intricate. Under Assumption 3.2, we categorize all units into six strata based on the joint potential outcome $\{I(0), I(1), Y(0), Y(1)\}$ from a counterfactual perspective (see Table 1). The descrip-

tive label of each stratum follows the motivating example introduced in Section 3.1.

For simplification, we denote the labels L for each stratum as $l_1, l_2, l_3, l_4, l_5, l_6$ and the return for assigning intervention to a unit as $r_1, r_2, r_3, r_4, r_5, r_6$. For $a, b, c \in \{0, 1\}$, let $\mu_{ab|c}(x) \equiv P(I = a, Y = b|T = c, x)$ and $\mu_{a|b}(x) \equiv P(Y = a|T = b, x)$. Also, define $e_j(x) \equiv P(L = l_j|x)$ for $j \in \{1, 2, 3, 4, 5, 6\}$ as the principal strata score of a unit belonging to a specific principal stratum l_j given covariate x .

3.3. Policy Evaluation

In this section, we extend the framework to account for differences in returns across strata. To learn an optimal policy from the observed data, we consider the utility function $u(t; l_j)$ that represents the utility of assigning an intervention decision t for a unit in stratum l_j . The return r_j can then be expressed as $u(1; l_j) - u(0; l_j)$ (Stefánsson, 2015; Bradley & Stefánsson, 2017). Subsequently, we define the value of a policy $\pi : \mathcal{X} \rightarrow \{0, 1\}$ as:

$$\begin{aligned} V(\pi) &= \mathbb{E} \left[\sum_{j=1}^6 \mathbb{I}\{L = l_j\} \{u(0; l_j)(1 - \pi(X)) + u(1; l_j)\pi(X)\} \right] \\ &= \mathbb{E} \left[\sum_{j=1}^6 e_j(X)\pi(X) \{u(1; l_j) - u(0; l_j)\} \right] \\ &+ \mathbb{E} \left[\sum_{j=1}^6 e_j(X)u(0; l_j) \right], \end{aligned}$$

where $\mathbb{I}(\cdot)$ is an indicator function. The value function consists of two parts. The latter part is irrelevant to the intervention assignment and represents the total utility in the natural state with no intervention, which does not affect learning of the optimal policy. Without loss of generality, we set this component to zero. The value of the former part is related to the intervention assignment, allowing for variations in utility under counterfactual outcomes.

Naturally, it is of interest to compare the values of two policies. We define the value loss $R_{e_2}(\pi, \psi)$ of policy π relative to another policy ψ as:

$$\begin{aligned} R_{e_2}(\pi, \psi) &\equiv V(\psi) - V(\pi) \\ &= \mathbb{E}[(\psi(X) - \pi(X)) \{s \cdot e_2(X) + c(X)\}], \end{aligned}$$

where $c(X) = \mu_{00|1}(X)r_1 + \mu_{01|1}(X)r_3 + \mu_{10|1}(X)r_4 + (\mu_{11|1}(X) - \mu_{11|0}(X))r_5 + (\mu_{11|0}(X) - \mu_{01|1}(X))r_6$ and $s = r_2 - r_3 - r_5 + r_6$. Detailed derivation of this equation is presented in Appendix D. Note that $R_{e_2}(\pi, \psi)$ represents the difference in overall returns between the two policies and is related to all principal strata scores e_j . However, based on the mathematical relationships between the principal strata scores e_j and the joint conditional probabilities $\mu_{ab|c}$ and $\mu_{a|b}$, we can simplify $R_{e_2}(\pi, \psi)$ into a form that involves

only e_2 . Without loss of generality, we select e_2 , which corresponds to users of Type B with higher values as discussed in Section 3.1, for presentational convenience. The same theoretical results hold when using alternative principal strata score (e_3, e_5 , or e_6), though the mathematical expressions differ (see Appendix D).

Notably, our policy value function and value loss function are designed for highly general settings, allowing arbitrary return values across different strata. However, $e_2(x)$ is always unidentifiable, which will be discussed further in Section 4.1. In this case, we define s as the “return indicator”, which serves as a central concept in establishing identifiability throughout our subsequent study.

4. The Minimax Approach

In this section, we first investigate partial identification of the principal strata score $e_2(x)$ and the value loss R_{e_2} . Then, we develop a minimax approach to derive the optimal policy by minimizing the worst-case value loss.

4.1. Partial Identification

A natural approach for policy learning is to seek the (unconstrained) optimal oracle policy π^o that maximizes $V(\cdot)$ or minimizes $R_{e_2}(\cdot)$:

$$\begin{aligned} \pi^o &\equiv \arg \max_{\pi} V(\pi) \\ &= \arg \min_{\pi} R_{e_2}(\pi, \psi) \\ &= \mathbb{I}\{s \cdot e_2(X) + c(X) \geq 0\}. \end{aligned}$$

However, a fundamental challenge arises due to a two-level non-identifiability: (i) the principal strata scores $e_2(X)$, $e_3(X)$, $e_5(X)$, and $e_6(X)$ are non-identifiable; and (ii) under general conditions where $s \neq 0$, both $V(\pi)$ and $R_{e_2}(\pi, \psi)$ are non-identifiable for policies π and ψ . We prove the non-identifiability and provide illustrative examples in Appendix D.

To address this issue, we partially identify the principal stratum score by deriving its sharp upper bound $U(x)$ and lower bounds $L(x)$:

Lemma 4.1.

$$\begin{aligned} L(x) &= \max\{0, \mu_{01|1}(x) - \mu_{11|0}(x)\} \\ &= \delta_l(x)(\mu_{01|1}(x) - \mu_{11|0}(x)), \\ U(x) &= \min\{\mu_{01|1}(x), \mu_{11|1}(x) - \mu_{11|0}(x)\} \\ &= \mu_{01|1}(x) + \delta_u(x)(\mu_{11|1}(x) - \mu_{11|0}(x)), \end{aligned}$$

where $\delta_l(x) = \mathbb{I}\{\mu_{01|1}(x) - \mu_{11|0}(x) \geq 0\}$ and $\delta_u(x) = \mathbb{I}\{\mu_{11|0}(x) - \mu_{11|1}(x) \geq 0\}$.

4.2. The Minimax Policy Relative to Different Alternative Policies

Since $e_2(X)$ is partially identifiable, we can get the bounds of $R_{e_2}(\pi, \psi)$ and consider the maximal value loss function $R_{\text{sup}}(\pi, \psi)$ to eliminate the impact of non-identifiability:

$$R_{\text{sup}}(\pi, \psi) = \max_{e_2(x) \in [L(x), U(x)]} R_{e_2}(\pi, \psi).$$

The maximal value loss function depends on the alternative policy ψ and is the maximum value across all possible values of $e_2(x)$. Given the alternative policy ψ and a budget ϵ , our minimax optimal policy is defined as:

$$\pi_{\psi}^* = \arg \min_{\pi} R_{\text{sup}}(\pi, \psi) \quad \text{s.t.} \quad \mathbb{E}[\pi(x)] \leq \epsilon.$$

This is a conservative optimal policy that minimizes the loss in the worst-case scenario.

We then use the minimax criterion to learn the conservative optimal policy relative to three alternative policies: (i) the “never-treat” policy $\pi^0(x) = 0$ for all x , (ii) the “always-treat” policy $\pi^1(x) = 1$ for all x , and (iii) the optimal oracle policy π^o that policymakers aim to achieve. These policies are standard benchmarks in the literature (Yata, 2021). The first two are non-personalized policies and the last one is the truly optimal policy.

We denote the (unconstrained) policies that minimize the worst-case losses relative to the three alternative policies as:

$$\begin{aligned} \pi_0^* &\equiv \arg \min_{\pi} R_{\text{sup}}(\pi, \pi^0), \\ \pi_1^* &\equiv \arg \min_{\pi} R_{\text{sup}}(\pi, \pi^1), \\ \pi_o^* &\equiv \arg \min_{\pi} R_{\text{sup}}(\pi, \pi^o). \end{aligned}$$

The following theorem presents the worst-case value loss relative to each of the three policies:

Theorem 4.2. *Let $\pi : X \rightarrow \{0, 1\}$ be a deterministic policy. For a comparison policy $\psi \in \{\pi^0, \pi^1, \pi^o\}$, the worst-case expected value loss of π relative to ψ is given by:*

$$R_{\text{sup}}(\pi, \psi) = C^{\psi} - \mathbb{E}[\pi(X)\{s \cdot b^{\psi}(X) + c(X)\}],$$

where C^{ψ} is a constant independent of π , and $b^{\psi}(X)$ is a key function depending on $\delta_l(\cdot)$, $\delta_u(\cdot)$, $\mu_{ab|c}(\cdot)$, and $\mu_{a|b}(\cdot)$.

The full statement of Theorem 4.2 is presented in Appendix B.1. Given the alternative policy ψ and a budget ϵ , we can then derive the optimal policy:

Proposition 4.3 (Optimal policy).

$$\pi_{\psi}^*(x_i) = \begin{cases} 1, & s \cdot b^{\psi}(x_i) + c(x_i) > \gamma(\epsilon; \psi), \\ 0, & s \cdot b^{\psi}(x_i) + c(x_i) \leq \gamma(\epsilon; \psi). \end{cases}$$

where $\gamma(\epsilon; \psi) > 0$ is a monotonically decreasing function of ϵ .

5. Policy Learning from Data

In this section, we present our optimal policy estimation algorithm and derive its theoretical guarantees. Proposition 4.3 demonstrates that identification of the optimal policy lies in estimation of parameters $b^{\psi}(x_i)$ and $c(x_i)$, both of which depend on the joint conditional probabilities of I and Y (i.e., $\mu_{ab|c}(x_i)$ and $\mu_{a|b}(x_i)$). Hence, the proposed learning framework consists of two steps: (i) estimating the conditional probabilities and (ii) estimating $R_{\text{sup}}(\pi, \psi)$ and then deriving the optimal policy $\hat{\pi}$ within a constrained policy class Π , where Π is a predefined collection of possible policies that can be selected.

5.1. Estimation of Conditional Probability

We propose three estimators – P-OR, P-IPS, and P-DR – to identify the joint conditional probability of I and Y . For illustration, we take $\mu_{01|1}(x)$ as an example, where the estimators of the other $\mu_{ab|c}(x)$ and $\mu_{a|b}(x)$ terms are similar. Specifically,

$$\begin{aligned} \mu_{01|1}(x) &= P(I = 0, Y = 1 | T = 1, x) \\ &= \mathbb{E}[\mathbb{I}(I = 0, Y = 1) | T = 1, x]. \end{aligned}$$

Since $\mathbb{I}(I = 0, Y = 1)$ is a binary variable, we can obtain an estimator $\hat{\mu}_{01|1}(\cdot)$ through regression, and the P-OR estimator of $\mu_{01|1}(x_i)$ is:

$$\hat{\mu}_{01|1}^{P-OR}(x_i) = \hat{\mu}_{01|1}(x_i)$$

Similar to the univariate case, we can obtain the P-IPS and P-DR estimators of $\mu_{01|1}(x_i)$ as:

$$\begin{aligned} \hat{\mu}_{01|1}^{P-IPS}(x_i) &= \frac{T_i(1 - I_i(1))Y_i(1)}{\hat{e}(x_i)}, \\ \hat{\mu}_{01|1}^{P-DR}(x_i) &= \hat{\mu}_{01|1}(x_i) + \frac{T_i \cdot \tilde{\mu}_{01|1}(x_i)}{\hat{e}(x_i)}. \end{aligned}$$

where $\hat{e}(x_i)$ is an estimator of propensity score $e(x_i)$ and $\tilde{\mu}_{01|1}(x_i) \equiv (1 - I_i(1))Y_i(1) - \hat{\mu}_{01|1}(x_i)$. Notably, under the consistency assumption, we have $T_i(1 - I_i(1))Y_i(1) = T_i(1 - I_i)Y_i$, which enables estimation using observed data. The P-DR estimator takes advantage of both P-OR and P-IPS estimators and possesses desirable statistical properties (detailed in Appendix B.2) that guarantee its estimation accuracy in large samples.

5.2. Learning the Optimal Policy

We estimate δ_l and δ_u using joint probability estimates, obtain $\hat{b}^{\psi}(\cdot)$ and $\hat{c}(\cdot)$ by plug-in estimation, and then derive the optimal policy $\hat{\pi}$ with budget ϵ :

$$\begin{aligned} \hat{\pi} &\in \arg \min_{\pi \in \Pi} \hat{R}_{\text{sup}}(\pi, \psi) \\ \text{s.t.} \quad &\frac{1}{n} \sum_{i=1}^n \hat{\pi}(x_i) \leq \epsilon, \end{aligned} \tag{1}$$

where

$$\hat{R}_{sup}(\pi, \psi) = -\frac{1}{n} \sum_{i=1}^n \pi(x_i) \{s \cdot \hat{b}^\psi(x_i) + \hat{c}(x_i)\}.$$

Here we present an example algorithm using the P-DR estimator with return indicator $s \geq 0$ and alternative policy π^0 .

Algorithm 1 Estimated the minimax policy $\hat{\pi}$ relative to the never-treat policy π^0 ($s \geq 0$).

Input: Policy classes Π and Δ_l

Output: Estimated minimax policy $\hat{\pi}$ relative to π^0

step 1 Find $\hat{\delta}_l$ by solving

$$\min_{\delta \in \Delta_l} \sum_{i=1}^n \delta(x_i) \left\{ \hat{\mu}_{1|0}^{P-DR}(x_i) - \hat{\mu}_{01|1}^{P-DR}(x_i) \right\}.$$

step 2 Estimate the key and cost functions, $\hat{b}^\psi$ and $\hat{c}^\psi$, using the P-DR estimates and $\hat{\delta}_l$ obtained in step 1.

step 3 Compute $\hat{R}_{sup}(\pi, \pi^0)$ using $\hat{b}^\psi$ and $\hat{c}$, and then find a policy

$$\hat{\pi} \in \arg \min_{\pi \in \Pi} \hat{R}_{sup}(\pi, \pi^0) \quad s.t. \quad \frac{1}{n} \sum_{i=1}^n \hat{\pi}(x_i) \leq \epsilon.$$

As detailed in Appendix C, the algorithms for other scenarios are similar to Algorithm 1.

5.3. Excess Worst-case Expected Value Loss

For a given alternative policy ψ , $R_{sup}(\pi_\psi^*, \psi)$ is the best possible expected value loss in the worst case. Therefore, the excess worst-case expected value loss $R_{sup}(\hat{\pi}, \psi) - R_{sup}(\pi_\psi^*, \psi)$ is an important and commonly used performance measurement for the learned optimal policy $\hat{\pi}$.

To present our theoretical results on the excess worst-case expected value loss, we first assess the quality of the estimated classifiers, $\hat{\delta}_l$ and $\hat{\delta}_u$, by their regrets:

$$R_l(\hat{\delta}_l) \equiv \mathbb{E} \left[\mathbb{I}\{\hat{\delta}_l(X) \neq \delta_l(X)\} |\mu_{01|1}(X) - \mu_{1|0}(X)| \right],$$

$$R_u(\hat{\delta}_u) \equiv \mathbb{E} \left[\mathbb{I}\{\hat{\delta}_u(X) \neq \delta_u(X)\} |\mu_{1|0}(X) - \mu_{11|1}(X)| \right].$$

Then, we measure the complexity of the policy class Π by its ability to overfit to noise via the population Rademacher complexity (Wainwright, 2019):

$$R_n(\Pi) \equiv \mathbb{E}_{X, \epsilon} \left[\sup_{\pi \in \Pi} \left| \frac{1}{n} \sum_{i=1}^n \epsilon_i \pi(X_i) \right| \right],$$

where ϵ_i 's are i.i.d. random variables with $\Pr(\epsilon_i = 1) = \Pr(\epsilon_i = -1) = \frac{1}{2}$, and the expectation is taken over both ϵ_i and X_i .

With these quantities, we derive the upper bound of the excess worst-case expected value loss relative to alternative policies $\psi = \pi^0$ and $\psi = \pi^1$. With data-splitting (Wager & Athey, 2018; Chernozhukov et al., 2017), we assume that the estimates have been obtained from a separate sample, and so can be treated as fixed for our finite-sample results.

Theorem 5.1. *Let $\hat{\pi}$ solve Eqn (1) with alternative policy $\psi = \pi^1$ (if $s < 0$) or $\psi = \pi^0$ (if $s \geq 0$) and no constraint. The estimates $\hat{e}$, $\hat{\mu}^{P-DR}$, and $\hat{\delta}_l$ are fitted on a separate sample. The excess worst-case expected value loss of $\hat{\pi}$ relative to π_ψ^* satisfies:*

$$\begin{aligned} & R_{sup}(\hat{\pi}, \psi) - R_{sup}(\pi_\psi^*, \psi) \\ & \leq U \cdot \left\{ \frac{12 + \eta}{\eta} (2R_n(\Pi) + \frac{t}{\sqrt{n}}) + \|\hat{\gamma}_0 - \gamma_0\|_2 \|\hat{\mu}_{1|0} - \mu_{1|0}\|_2 \right. \\ & \quad + \|\hat{\gamma}_1 - \gamma_1\|_2 (\|\hat{\mu}_{00|1} - \mu_{00|1}\|_2 + \|\hat{\mu}_{01|1} - \mu_{01|1}\|_2 \\ & \quad \left. + \|\hat{\mu}_{10|1} - \mu_{10|1}\|_2) \right\} + 2|s| \cdot (R_l(\hat{\delta}_l) + \frac{t}{2\sqrt{n}}), \end{aligned}$$

with a probability of at least $1 - 2 \exp\left(-\frac{t^2}{2}\right)$, where η is the overlap parameter in Assumption 3.1, U is a constant depending on the return values, $\|\hat{\gamma}_1 - \gamma_1\|_2^2 \equiv \mathbb{E} \left[\left\{ \frac{T}{e(X)} - \frac{T}{\hat{e}(X)} \right\}^2 \right]$, $\|\hat{\gamma}_0 - \gamma_0\|_2^2 \equiv \mathbb{E} \left[\left\{ \frac{1-T}{1-e(X)} - \frac{1-T}{1-\hat{e}(X)} \right\}^2 \right]$, and $\|\hat{\mu}_{01|1}(\cdot) - \mu_{01|1}(\cdot)\|_2^2 \equiv \mathbb{E} \left[\{\hat{\mu}_{01|1}(X) - \mu_{01|1}(X)\}^2 \right]$.

As detailed in Appendix B.3, the results for $\psi = \pi^1$ (if $s \geq 0$) or $\psi = \pi^0$ (if $s < 0$) are similar to those in Theorem 5.1. Building upon Theorem 5.1, we further derive the upper bound of the excess worst-case expected value loss relative to the oracle policy $\psi = \pi^o$.

Theorem 5.2. *Let $\hat{\pi}_o$ solve Eqn 1 with the alternative policy $\psi = \pi^o$ and no constraint. The estimates $\hat{e}$, $\hat{\mu}^{P-DR}$, $\hat{\delta}_l$, $\hat{\delta}_u$, and estimated policies $\hat{\pi}_0$, $\hat{\pi}_1$ are fitted on a separate sample. The excess worst-case expected value loss of $\hat{\pi}_o$ relative to π_o^* satisfies:*

$$\begin{aligned} & R_{sup}(\hat{\pi}_o, \psi) - R_{sup}(\pi_o^*, \psi) \\ & \leq U \cdot \left\{ \frac{12 + \eta}{\eta} (2R_n(\Pi) + \frac{t}{\sqrt{n}}) + \|\hat{\gamma}_0 - \gamma_0\|_2 \|\hat{\mu}_{1|0} - \mu_{1|0}\|_2 \right. \\ & \quad + \|\hat{\gamma}_1 - \gamma_1\|_2 (\|\hat{\mu}_{00|1} - \mu_{00|1}\|_2 + \|\hat{\mu}_{01|1} - \mu_{01|1}\|_2 \\ & \quad \left. + \|\hat{\mu}_{10|1} - \mu_{10|1}\|_2) \right\} + 2|s| \cdot (R_l(\hat{\delta}_l) + R_u(\hat{\delta}_u) + \frac{t}{2\sqrt{n}}) \\ & \quad + 4 \sum_{w=0}^1 \{R_{sup}(\hat{\pi}_w, \pi_w) - R_{sup}(\pi_w^*, \pi_w)\}, \end{aligned}$$

with a probability of at least $1 - 2 \exp\left(-\frac{t^2}{2}\right)$, where U is a constant depending on the return values.

Theorem 5.1 and 5.2 explicitly show that the overall convergence rate of regret is determined by four key components: (i) the Rademacher complexity $R_n(\Pi)$, (ii) the estimation errors of propensity scores $\|\hat{\gamma}_a - \gamma_a\|_2$, (iii) the regression estimation errors $(\|\hat{\mu}_{ab|c} - \mu_{ab|c}\|_2)$, and (iv) the classification

errors $R_t(\hat{\delta}_t)$ and $R_u(\hat{\delta}_u)$. There exists extensive literature examining the properties of these components (Kitagawa & Tetenov, 2018; Bickel & Doksum, 2015). When the policy class is of low complexity and the nuisance parameters are estimated with sufficient accuracy, the maximal value loss of our estimated policy converges rapidly to the true value as sample size increases. For instance, under standard conditions, the regret can achieve an $O(\frac{1}{\sqrt{n}})$ overall convergence rate: using Support Vector Machine (SVM) as the learner (policy class), the Rademacher complexity achieves $O(\frac{1}{\sqrt{n}})$ (Mohri et al., 2018), and the remaining three components can achieve similar rates (Athey & Wager, 2021).

In the constrained scenario with a budget ϵ , analogous results can be obtained. The constraint affects only the policy class Π and does not influence the estimation of other nuisance parameters. Specifically, it reduces the Rademacher complexity $R_n(\Pi)$, thereby leading to tighter regret upper bounds than those presented in Theorems 5.1 and 5.2.

6. Experiments

6.1. Experimental Setup

Dataset and Preprocessing. To further evaluate the performance of the proposed learning framework, we conduct extensive experiments on three publicly available datasets ML-1M², Jester³, and KuaiRec⁴. In this section, we present the operations and experimental results for ML-1M and Jester. Analysis of KuaiRec, which shows consistent results, is presented in Appendix G. ML-1M is a movie rating dataset that contains user-movie information with movie ratings as the outcome of interest. Jester is a joke rating dataset containing user-joke interactions with continuous ratings from -10 to $+10$ as the feedback metric.

To better explain the practical relevance of our method, consider the following real-world-inspired scenario with the ML-1M movie rating data: (i) incentive T : the production company invites users to attend a live “question and answer” roadshow (1 = invited, 0 = not invited); (ii) first-stage outcome I : whether the user attends the roadshow (1 = attend, 0 = do not attend); and (iii) second-stage outcome Y : whether the user rates the movie highly (1 = high rating, 0 = low rating). While attending the roadshow events ($I(1) = 1$) tends to promote high ratings ($Y(1) = 1$), organizing such events incurs substantial costs, and the company can only accommodate a limited number of attendees with a budget. Our method emphasizes identifying and targeting a key subgroup: “No Claim-Incentive Buyers” – users who interpret the invitation as a friendly signal and give high ratings without actually attending the event. By prioritizing

this group, the policy achieves desirable outcomes while reducing operational costs.

Following previous studies (Liu et al., 2020; 2019; Wu et al., 2016a;b), we binarize the observed ratings in ML-1M by setting $Y = 1$ for ratings ≥ 0 and $Y = 0$ otherwise, assuming $I(1) = 1$ for all observations. Similarly, we binarize the observed ratings in Jester by setting $Y = 1$ for ratings ≥ 0 and $Y = 0$ otherwise, assuming $I(1) = 1$ for all observations. Next, we perform counterfactual strata labeling. Notably, all the ($I(1) = 1, Y(1) = 0$) units must be “Incentive Claim - Never Buyers” in Table 1. For the remaining units with other user-movie/joke interactions, which are not observed (assuming $I(0) = 0$), we pre-train Neural Collaborative Filtering (NCF) (He et al., 2017) models to generate the predicted outcome Y (i.e., ratings). Then, for the ($I(1) = 1, Y(1) = 1$) units, we treat 80% with the highest predicted outcome as “Incentive Claim - Incentive Buyers” and the remaining 20% as “Incentive Claim - Always Buyers”. For units with $I(1) = 0$, we label those in the top 40% of the predicted outcome as “No Claim - Incentive Buyers”. For the remaining units with $I(1) = 0$, we label half with the higher predicted outcomes as “No Claim - Always Buyers” and the rest as “No Claim - Never Buyers”.

Experimental Details. We set the strata-specific returns as $\{0, 1.2, 0, 0, 1 - C, -0.2 - C\}$ for these three datasets, where C denotes the intervention cost. We compare the proposed method with two widely adopted approaches: (i) the association-based naive method, which allocates incentives based on predicted outcomes (i.e., prioritizing units with higher Y) and (ii) the uplift modeling methods, which determine the incentive assignment using the OR, IPS, and DR estimators (i.e., prioritizing units with higher CATE). The detailed implementation of both approaches can be found in Appendix G. NCF is used for both the regression model and the propensity model.

Additional Experiments. In Appendix G, we conduct extensive additional experiments across various settings to further validate the desirable properties of our method. These include: employing a more flexible specification of principal stratum returns; introducing mild violations of the monotonicity assumption in Assumption 3.2; adjusting the proportions of principal strata and the configuration of strata labeling in the data-generating process; changing the base learner of the predictive model; considering scenarios with completely unconstrained interventions; and comparing our results with methods that rely on the exclusion assumption (Li et al., 2023b).

6.2. Experimental Results

Overall Performance. Table 2 presents the number of interventions administered to different strata and the overall

²<https://grouplens.org/datasets/movielens/1M/>

³<https://eigentaste.berkeley.edu/dataset/>

⁴<https://github.com/chongminggao/KuaiRec>

Table 2. Experimental results with $r_1, r_2, r_3, r_4, r_5, r_6$ set as $[0, 1.2, 0, 0, 0.6, -0.6]$.

COUPON		ML-1M						
METHODS	Y ₀₀₀₀	Y ₀₀₀₁	Y ₀₀₁₁	Y ₀₁₀₀	Y ₀₁₀₁	Y ₀₁₁₁	RETURN	RI
NAIVE	31,785	17,222	236,563	50,931	46,013	9,341	42,670	-
OR	141,626	42,201	17,582	62,836	114,529	13,081	111,510	-
P-OR	39,773	67,295	108,406	54,187	107,359	14,835	136,268	22.2%
IPS	164,920	41,969	10,561	51,533	113,317	9,555	112,620	-
P-IPS	114,361	66,331	1130	35,320	116,422	3,449	147,381	30.8%
DR	151,877	37,393	16,344	60,619	112,276	13,346	104,230	-
P-DR	115,130	68,697	3,672	39,840	117,821	4,479	150,442	44.3%

COUPON		JESTER						
METHODS	Y ₀₀₀₀	Y ₀₀₀₁	Y ₀₀₁₁	Y ₀₁₀₀	Y ₀₁₀₁	Y ₀₁₁₁	RETURN	RI
NAIVE	10,811	13,822	2,423	77,285	100,844	6,305	77,310	–
OR	42	20,148	9	47,119	137,759	6,414	102,985	–
P-OR	26,550	106,534	10,450	22,924	39,252	5,781	147,923	43.6%
IPS	112	13,009	3	57,555	127,026	13,786	83,555	–
P-IPS	2,770	74,942	197	41,605	80,456	11,521	131,291	57.1%
DR	138	18,130	15	48,675	137,400	7,133	99,916	–
P-DR	2,781	75,080	198	41,475	80,453	11,504	131,465	31.6%

return by policies, as well as the percentage of return improvement (RI) comparing our methods to the uplift modeling methods. The analysis reveals significant findings. First, among all methods evaluated, the naive method performs the worst, while all the uplift modeling methods, i.e., OR, IPS, and DR, show significant improvements. This validates the importance of estimating causal effects for personalized intervention policies. Next, all the proposed methods, i.e., P-OR, P-IPS, and P-DR, show significant improvement compared to the uplift modeling methods. A direct reason of the improvement is that the proposed approach selects more units from the Y_{0001} stratum (i.e., “No Claim - Incentive Buyers”) with higher returns. This indicates that the proposed approach can successfully identify individuals belonging to each of the six counterfactual strata, whereas the uplift modeling methods can only identify a combinative group of “No Claim - Incentive Buyers” and “Incentive Claim - Incentive Buyers”. Although we adopt a conservative minimax strategy, the improvement is significant.

Effects of Varying Cost. We further study the effect of intervention (coupon) cost on policy performance (see Figure 2), where we compare the naive method, DR of the uplift modeling methods, and P-DR of the proposed methods. As cost increases, the naive method consistently performs the worst, while the proposed method consistently outperforms the others and exhibits the slowest rate of decline. This is because the return for “No Claim - Incentive Buyers” is not affected by changes in cost, and only our method is capable of capturing this stratum.

Effects of Varying Budget. We also study the effect of budget on policy performance (see Figure 3), where the

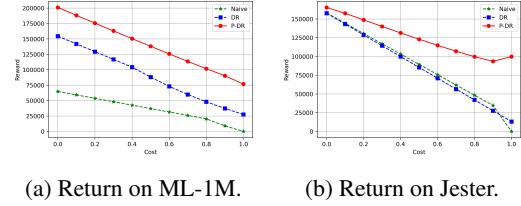


Figure 2. Effects of varying cost.

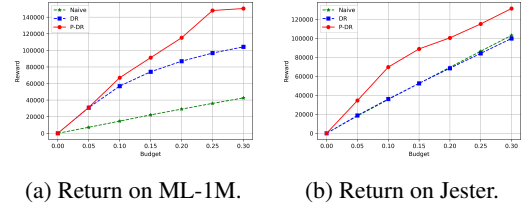


Figure 3. Effects of varying budget.

same methods are adopted as in Figure 2. As budget increases, the naive method consistently performs the worst, while the proposed method consistently outperforms the others and exhibits the fastest rate of improvement. This is because under limited budget, administering more interventions to “No Claim - Incentive Buyers” yields higher returns. When the budget exceeds 0.3, the returns of the DR and P-DR methods increase only marginally, which is due to the limited proportion of individuals with positive “intervention effects. Therefore, results for budget exceeds 0.3 are omitted in Figure 3.

7. Conclusion

This paper studies the problem of optimal policy learning through counterfactual individual responses. We propose a minimax approach, introduce joint probability estimators, and prove the theoretical upper bound of the estimation error. One limitation of the work is that it has not yet been applied to real-world scenarios. Another limitation is that some restrictive conditions and assumptions within the principal stratification framework have not been sufficiently relaxed. For example, our work can be extended to scenarios involving multiple treatments and multi-stage outcomes. Future efforts should focus on extending our methods and theories by relaxing these conditions and assumptions and applying them to practical scenarios.

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Impact Statement

This paper introduces a minimax approach within a counterfactual principal strata framework, aimed to learn optimal intervention policies while addressing challenges such as partial identification and various principal returns. This framework holds significant potential for diverse applications. In personalized services, such as e-commerce platforms and recommendation systems, it can be employed to design customized incentive strategies aimed at increasing user conversion rates and driving revenue growth. Additionally, it can be extended to critical sectors like healthcare and education, where it can guide the more effective allocation of vital resources to improve outcomes. By improving data-driven decision-making, this framework not only enhances operational efficiency across these fields but also contributes to the conservation of societal resources, ultimately maximizing social welfare.

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A. Difference form the IV Framework

Although the notation of the IV framework (Schur et al., 2026) may appear similar to that used in our work, the research problems addressed by the two approaches are fundamentally distinct.

First, in the standard IV setting, potential outcomes of the outcome variable Y are indexed by the intermediate variable I , i.e., $Y(1) \equiv Y(I = 1)$. In contrast, our frame work defines the joint potential outcomes (Y, I) with respect to the treatment variable T , i.e., $Y(1), I(1)$ jointly represent the two-stage potential outcomes of interest under treatment assignment $T = 1$.

Second, existing works on IV primarily focus on the causal effect, meaning that IV methods can solely help to identify/estimate certain marginal distributions of potential outcomes (e.g., the distribution of $Y(1)$, the distribution of $(Y(1), I(1))$, and the point value of $E(Y(1) - Y(0))$), which may yield relatively sharp or point-identified results. However, IV methods can not identify/estimate the joint distribution of potential outcomes (i.e., the distribution of $(I(0), I(1), Y(0), Y(1))$). This limitation stems from the fundamental identification problem in causal inference: we can never simultaneously observe the potential outcomes with and without interventions for the same unit (Holland, 1986). Classic literature about IV such as Balke-Pearl bounds (Balke & Pearl, 1997) use IV to partially identify such causal effects. In contrast, our work is not centered on the causal effect of I on Y , but rather on how different joint counterfactual outcomes of I and Y lead to different returns, which the IV framework is not designed to address. Specifically, we aim to (partially) identify these joint counterfactual outcomes (i.e., principal strata) and learn an optimal minimax policy to maximize total return.

Overall, our approach is more appropriate for addressing real-world problems raised in this paper, in which the counterfactual outcomes of I are also of major interest. Consider an advertising scenario where Company A delivers software ads via Platform B: Treatment T represents ad exposure, with first-stage outcome I capturing ad clicks and second-stage outcome Y measuring software downloads, where clicks ($I = 1$) positively influence downloads ($Y = 1$). Here, I is important to study because Platform B generally charges Company A based on click-throughs ($I = 1$) in practice. However, IV-based approaches do not account for potential outcomes of I . Other examples include e-commerce incentives to promote purchases, preventive healthcare reminders to boost treatment adherence, among others — for all of which subject engagement with the intervention ($I = 1$) incur costs.

B. Additional Theoretical Results

B.1. Full Statement of Theorem 4.2

Let $\pi : X \rightarrow \{0, 1\}$ be a deterministic policy. For comparison policy $\psi \in \{\pi^0, \pi^1, \pi^o\}$, the worst-case expected value loss of π relative to ψ is given by:

$$R_{\text{sup}}(\pi, \psi) = C^\psi - \mathbb{E}[\pi(X)\{s \cdot b^\psi(X) + c(X)\}],$$

where C^ψ is a constant independent of π , and $b^\psi(X)$ is a key function depends on $\delta_l(\cdot)$, $\delta_u(\cdot)$ and $\mu(\cdot)$.

For $s \geq 0$,

$$\begin{aligned} b^{\pi^0}(X) &= \delta_l(X)(\mu_{01|1}(X) - \mu_{1|0}(X)), \\ b^{\pi^1}(X) &= \mu_{01|1}(X) + \delta_u(X)(\mu_{11|1}(X) - \mu_{1|0}(X)), \\ b^{\pi^o}(X) &= \pi_0^*(X)\{\mu_{01|1}(X) + \delta_u(X)(\mu_{11|1}(X) - \mu_{1|0}(X))\} \\ &\quad + (1 - \pi_1^*(X))\{\delta_l(X)(\mu_{01|1}(X) - \mu_{1|0}(X))\} \\ &\quad + (1 - \pi_0^*(X))\pi_1^*(X)(\mu_{01|1}(X) + \delta_u(X)\{\mu_{11|1}(X) - \mu_{1|0}(X)\}) \\ &\quad + \delta_l(X)\{\mu_{01|1}(X) - \mu_{1|0}(X)\}. \end{aligned}$$

For $s < 0$,

$$\begin{aligned} b^{\pi^0}(X) &= \mu_{01|1}(X) + \delta_u(X)(\mu_{11|1}(X) - \mu_{1|0}(X)), \\ b^{\pi^1}(X) &= \delta_l(X)(\mu_{01|1}(X) - \mu_{1|0}(X)), \\ b^{\pi^0}(X) &= \pi_0^*(X)\{\mu_{01|1}(X) + \delta_l(X)(\mu_{01|1}(X) - \mu_{1|0}(X))\} \\ &\quad + (1 - \pi_1^*(X))\{\mu_{01|1}(X) + \delta_u(X)(\mu_{11|1}(X) - \mu_{1|0}(X))\} \\ &\quad + (1 - \pi_0^*(X))\pi_1^*(X)(\mu_{01|1}(X) + \delta_u(X)\{\mu_{11|1}(X) - \mu_{1|0}(X)\}) \\ &\quad + \delta_l(X)\{\mu_{01|1}(X) - \mu_{1|0}(X)\}, \end{aligned}$$

where $\pi_0^*(X) = \mathbb{I}\{s \cdot b^{\pi^0}(X) + c(X) \geq 0\}$, $\pi_1^*(X) = \mathbb{I}\{s \cdot b^{\pi^1}(X) + c(X) \geq 0\}$.

B.2. Statistical properties of P-DR Estimators

Theorem B.1 (Bias and Double Robustness of the P-DR Estimator). *(a) Given the imputed outcome $\hat{\mu}_{01|1}(x_i)$ and the learned propensity score $\hat{e}(x_i) \in (0, 1)$ for all units, the bias of the P-DR estimator is:*

$$\text{Bias}(\hat{\mu}_{01|1}^{P-DR}(x_i)) = \frac{|(e(x_i) - \hat{e}(x_i))\tilde{\mu}_{01|1}(x_i)|}{\hat{e}(x_i)}.$$

(b) The P-DR estimator is unbiased when either the imputed outcome $\hat{\mu}_{01|1}(x_i)$ or the learned propensities $\hat{e}(x_i) \in (0, 1)$ are accurate for all units.

Theorem B.2 (Variance of the P-DR Estimator). *Given the imputed outcome $\hat{\mu}_{01|1}(x_i)$ and the learned propensity score $\hat{e}(x_i) \in (0, 1)$ for all units, the variance of the P-DR estimator is*

$$\text{Var}(\hat{\mu}_{01|1}^{P-DR}(x_i)) = e(x_i)(1 - e(x_i)) \left[\frac{\tilde{\mu}_{01|1}(x_i)}{\hat{e}(x_i)} \right]^2.$$

Notably, when the outcome regression models are approximately accurate, i.e., $(1 - I_i(1))Y_i(1) - \hat{\mu}_{01|1}(x_i) \approx 0$, the P-DR estimator has a lower variance than the P-IPS.

Let $\hat{\mu}_{01|1}^{P-DR}$ be the average predicted joint probability of $\{I = 0, Y = 1|T = 1\}$ using the P-DR estimator over all user-item pairs

$$\hat{\mu}_{01|1}^{P-DR} = \frac{1}{n} \sum_{i=1}^n \hat{\mu}_{01|1}^{P-DR}(x_i).$$

We further show the tail bound of P-DR estimator in Theorem B.3:

Theorem B.3 (Tail Bound of the P-DR Estimator). *Given the imputed outcome $\hat{\mu}_{01|1}(x_i)$ and the learned propensity score $\hat{e}(x_i) \in (0, 1)$, with probability $1 - \eta$, the deviation of the P-DR estimator from its expectation has the following tail bound:*

$$\left| \hat{\mu}_{01|1}^{P-DR} - \mathbb{E}(\hat{\mu}_{01|1}^{P-DR}) \right| \leq \sqrt{\frac{\log\left(\frac{2}{\eta}\right)}{2n^2} \sum_{i=1}^n \left[\frac{\tilde{\mu}_{01|1}(x_i)}{\hat{e}(x_i)} \right]^2}.$$

Theorem B.1 and Theorem B.3 demonstrate that if either the imputed outcomes or the learned propensities are approximately accurate, the predicted joint probability from the P-DR estimator will converge to the true value as the sample size increases.

B.3. Supplementary Theorem in Section 5.3

Theorem B.4. *Let $\hat{\pi}$ solve Eqn (1) with alternative policy $\psi = \pi^1$ (if $s \geq 0$) or $\psi = \pi^0$ (if $s < 0$) and no constraint. The estimates $\hat{e}$, $\hat{\mu}^{P-DR}$, and $\hat{\delta}_u$ are fitted on a separate sample. The excess worst-case expected value loss of $\hat{\pi}$ relative to π_ψ^**

satisfies:

$$\begin{aligned}
 R_{sup}(\hat{\pi}, \psi) - R_{sup}(\pi_{\psi}^*, \psi) &\leq U \times \left\{ \frac{12 + \eta}{\eta} \times (2R_n(\Pi) + \frac{t}{\sqrt{n}}) + \|\hat{\gamma}_0 - \gamma_0\|_2 \|\hat{\mu}_{1|0} - \mu_{1|0}\|_2 \right. \\
 &\quad \left. + \|\hat{\gamma}_1 - \gamma_1\|_2 (\|\hat{\mu}_{00|1} - \mu_{00|1}\|_2 + \|\hat{\mu}_{01|1} - \mu_{01|1}\|_2 + \|\hat{\mu}_{10|1} - \mu_{10|1}\|_2) \right\} \\
 &\quad + 2|s| \times (R_u(\hat{\delta}_u) + \frac{t}{2\sqrt{n}}),
 \end{aligned}$$

with a probability of at least $1 - 2 \exp\left(-\frac{t^2}{2}\right)$, where η is the overlap parameter in Assumption 3.1, U is a constant depending on the return values, $\|\hat{\gamma}_1 - \gamma_1\|_2^2 \equiv \mathbb{E} \left[\left\{ \frac{T}{e(X)} - \frac{T}{\bar{e}(X)} \right\}^2 \right]$, $\|\hat{\gamma}_0 - \gamma_0\|_2^2 \equiv \mathbb{E} \left[\left\{ \frac{1-T}{1-e(X)} - \frac{1-T}{1-\bar{e}(X)} \right\}^2 \right]$, and $\|\hat{\mu}_{01|1}(\cdot) - \mu_{01|1}(\cdot)\|_2^2 \equiv \mathbb{E} \left[\{\hat{\mu}_{01|1}(X) - \mu_{01|1}(X)\}^2 \right]$.

C. Supplementary Algorithms in Section 5.2

Algorithm B.1: Estimate the minimax policy $\hat{\pi}$ relative to the always-treat policy π^1 .

Input: Policy classes Π , Δ_l and Δ_u

Output: Estimate minimax policy $\hat{\pi}$ relative to π^1

step 1 Find $\hat{\delta}_l$ and $\hat{\delta}_u$ by solving

$$\begin{aligned} \min_{\delta_l \in \Delta_l} & -\frac{1}{n} \sum_{i=1}^n \delta_l(x_i) \left\{ \hat{\mu}_{01|1}^{P-DR}(x_i) - \hat{\mu}_{1|0}^{P-DR}(x_i) \right\}, \\ \min_{\delta_u \in \Delta_u} & -\frac{1}{n} \sum_{i=1}^n \delta_u(x_i) \left\{ \hat{\mu}_{11|1}^{P-DR}(x_i) - \hat{\mu}_{1|0}^{P-DR}(x_i) \right\}. \end{aligned}$$

step 2 Compute the key and cost functions

$$\begin{aligned} \hat{c}(x_i) &= \hat{\mu}_{00|1}^{P-DR}(x_i)r_1 + \hat{\mu}_{01|1}^{P-DR}(x_i)r_3 + \hat{\mu}_{10|1}^{P-DR}(x_i)r_4 \\ &+ (\hat{\mu}_{11|1}^{P-DR}(x_i) - \hat{\mu}_{1|0}^{P-DR}(x_i))r_5 + (\hat{\mu}_{1|0}^{P-DR}(x_i) - \hat{\mu}_{01|1}^{P-DR}(x_i))r_6. \end{aligned}$$

if $s < 0$

$$\hat{b}^{\pi^1}(x_i) = \hat{\delta}_l(x_i)(\hat{\mu}_{01|1}^{P-DR}(x_i) - \hat{\mu}_{1|0}^{P-DR}(x_i)).$$

else

$$\hat{b}^{\pi^1}(x_i) = \hat{\mu}_{01|1}(x_i) + \hat{\delta}_u(x_i)(\hat{\mu}_{11|1}^{P-DR}(x_i) - \hat{\mu}_{1|0}^{P-DR}(x_i)).$$

step 3 Compute

$$\hat{R}_{sup}(\pi, \pi^1) = -\frac{1}{n} \sum_{i=1}^n \pi(x_i) \{s \cdot \hat{b}^{\pi^1}(x_i) + \hat{c}(x_i)\},$$

and then find a policy $\hat{\pi} \in \arg \min_{\pi \in \Pi} \hat{R}_{sup}(\pi, \pi^1) \quad s.t. \frac{1}{n} \sum_{i=1}^n \hat{\pi}(x_i) \leq \epsilon$.

Algorithm B.2: Estimate the minimax policy $\hat{\pi}$ relative to the oracle policy π^o .

Input: Policy classes Π , Δ_l and Δ_u

Output: Estimate minimax policy $\hat{\pi}$ relative to π^o

step 1 Find $\hat{\delta}_l$ and $\hat{\delta}_u$ by solving

$$\begin{aligned} \min_{\delta_l \in \Delta_l} & -\frac{1}{n} \sum_{i=1}^n \delta_l(x_i) \left\{ \hat{\mu}_{01|1}^{P-DR}(x_i) - \hat{\mu}_{1|0}^{P-DR}(x_i) \right\}, \\ \min_{\delta_u \in \Delta_u} & -\frac{1}{n} \sum_{i=1}^n \delta_u(x_i) \left\{ \hat{\mu}_{11|1}^{P-DR}(x_i) - \hat{\mu}_{1|0}^{P-DR}(x_i) \right\}. \end{aligned}$$

step 2 Estimate minimax policy $\hat{\pi}_1$ relative to π^1 and $\hat{\pi}_0$ relative to π_0 .

step 3 Compute the key and cost functions

$$\begin{aligned} \hat{c}(x_i) &= \hat{\mu}_{00|1}^{P-DR}(x_i)r_1 + \hat{\mu}_{01|1}^{P-DR}(x_i)r_3 + \hat{\mu}_{10|1}^{P-DR}(x_i)r_4 \\ &+ (\hat{\mu}_{11|1}^{P-DR}(x_i) - \hat{\mu}_{1|0}^{P-DR}(x_i))r_5 + (\hat{\mu}_{1|0}^{P-DR}(x_i) - \hat{\mu}_{01|1}^{P-DR}(x_i))r_6. \end{aligned}$$

Algorithm B.2 (continued): Estimate the minimax policy $\hat{\pi}$ relative to the oracle policy π^o .

if $s < 0$

$$\begin{aligned} b^{\pi^o}(x_i) = & \hat{\pi}_0(x_i) \{ \hat{\mu}_{01|1}^{P-DR}(x_i) + \delta_l(x_i) (\hat{\mu}_{01|1}^{P-DR}(x_i) - \hat{\mu}_{1|0}^{P-DR}(x_i)) \} \\ & + (1 - \hat{\pi}_1(x_i)) \{ \hat{\mu}_{01|1}^{P-DR}(x_i) + \delta_u(x_i) (\hat{\mu}_{11|1}^{P-DR}(x_i) - \hat{\mu}_{1|0}^{P-DR}(x_i)) \} \\ & + (1 - \hat{\pi}_0(x_i)) \hat{\pi}_1(x_i) \{ \hat{\mu}_{01|1}^{P-DR}(x_i) + \delta_u(x_i) \{ \hat{\mu}_{11|1}^{P-DR}(x_i) - \hat{\mu}_{1|0}^{P-DR}(x_i) \} \\ & + \delta_l(x_i) \{ \hat{\mu}_{01|1}^{P-DR}(x_i) - \hat{\mu}_{1|0}^{P-DR}(x_i) \} \}. \end{aligned}$$

else

$$\begin{aligned} b^{\pi^o}(x_i) = & \hat{\pi}_0(x_i) \{ \hat{\mu}_{01|1}^{P-DR}(x_i) + \delta_u(x_i) (\hat{\mu}_{11|1}^{P-DR}(x_i) - \hat{\mu}_{1|0}^{P-DR}(x_i)) \} \\ & + (1 - \hat{\pi}_1(x_i)) \{ \delta_l(x_i) (\hat{\mu}_{01|1}^{P-DR}(x_i) - \hat{\mu}_{1|0}^{P-DR}(x_i)) \} \\ & + (1 - \hat{\pi}_0(x_i)) \hat{\pi}_1(x_i) \{ \hat{\mu}_{01|1}^{P-DR}(x_i) + \delta_u(x_i) \{ \hat{\mu}_{11|1}^{P-DR}(x_i) - \hat{\mu}_{1|0}^{P-DR}(x_i) \} \\ & + \delta_l(x_i) \{ \hat{\mu}_{01|1}^{P-DR}(x_i) - \hat{\mu}_{1|0}^{P-DR}(x_i) \} \}. \end{aligned}$$

step 4 Compute

$$\hat{R}_{sup}(\pi, \pi^o) = -\frac{1}{n} \sum_{i=1}^n \pi(x_i) \{ s \cdot \hat{b}^{\pi^o}(x_i) + \hat{c}(x_i) \},$$

and then find a policy $\hat{\pi} \in \arg \min_{\pi \in \Pi} \hat{R}_{sup}(\pi, \pi^o)$ s.t. $\frac{1}{n} \sum_{i=1}^n \hat{\pi}(x_i) \leq \epsilon$.

D. Proofs in Section 3 and Section 4

Derivation of the value loss. The value loss $R_{e_2}(\pi, \psi)$ of policy π relative to another policy ψ is

$$R_{e_2}(\pi, \psi) \equiv V(\psi) - V(\pi) = \mathbb{E}[(\psi(X) - \pi(X)) \{s \cdot e_2(X) + c(X)\}].$$

proof. First, notice that the value of policy π is

$$V(\pi) = \mathbb{E} \left[\sum_{j=1}^6 e_j(X) \pi(X) \{u(1; l_j) - u(0; l_j)\} \right] + \mathbb{E} \left[\sum_{j=1}^6 e_j(X) u(0; l_j) \right].$$

The first term can be written as

$$(*) = \mathbb{E} \left[\sum_{j=1}^6 e_j(X) \pi(X) \{u(1; l_j) - u(0; l_j)\} \right] = \mathbb{E} \left[\pi(X) \left\{ \sum_{j=1}^6 e_j(X) r_j \right\} \right].$$

Solving the equations in Section 4.1, we can derive

$$\begin{aligned} e_1(X) &= \mu_{00|1}(X), \quad e_3(X) = \mu_{01|1}(X) - e_2(X), \quad e_4(X) = \mu_{10|1}(X), \\ e_5(X) &= \mu_{11|1}(X) - \mu_{1|0}(X) + \mu_{01|1}(X) - e_2(X), \quad e_6(X) = \mu_{1|0}(X) - \mu_{01|1}(X) + e_2(X). \end{aligned}$$

Then we can further obtain

$$(*) = \mathbb{E}[\pi(X) \{s \cdot e_2(X) + c(X)\}],$$

where $c(X) = \mu_{00|1}(X)r_1 + \mu_{01|1}(X)r_3 + \mu_{10|1}(X)r_4 + (\mu_{1|1}(X) - \mu_{1|0}(X))r_5 + (\mu_{1|0}(X) - \mu_{01|1}(X))r_6$ and $s = r_2 - r_3 - r_5 + r_6$. So the value loss of policy π relative to policy ψ is

$$R_{e_2}(\pi, \psi) \equiv V(\psi) - V(\pi) = \mathbb{E}[(\psi(X) - \pi(X)) \{s \cdot e_2(X) + c(X)\}].$$

□

Non-identifiability: (i) the principal strata scores $e_2(X)$, $e_3(X)$, $e_5(X)$ and $e_6(X)$ are non-identifiable, and (ii) under general conditions where $s \neq 0$, $V(\pi)$ and $R_{e_2}(\pi, \psi)$ are non-identifiable for policies π and ψ .

Proof. Considering the relationship between the observed data $\mu_{ab|c}(x)$, $\mu_{a|b}(x)$ and the principal score $e_j(x)$, we have:

$$\begin{aligned}\mu_{00|1}(X) &= e_1(X), \\ \mu_{01|1}(X) &= e_2(X) + e_3(X), \\ \mu_{10|1}(X) &= e_4(X), \\ \mu_{11|1}(X) &= e_5(X) + e_6(X), \\ \mu_{1|0}(X) &= e_3(X) + e_6(X), \\ \mu_{0|0}(X) &= e_1(X) + e_2(X) + e_4(X) + e_5(X).\end{aligned}$$

It is noted that two of the equations are redundant, as there are three potential equations relating the observed data and the principal scores:

$$\begin{aligned}\mu_{00|1}(X) + \mu_{01|1}(X) + \mu_{10|1}(X) + \mu_{11|1}(X) &= 1, \\ \mu_{1|0}(X) + \mu_{0|0}(X) &= 1, \\ e_1(X) + e_2(X) + e_3(X) + e_4(X) + e_5(X) + e_6(X) &= 1.\end{aligned}$$

Ultimately, we have five valid equations but six unknown strata probabilities, from which we can only identify $e_1(x)$ and $e_4(x)$ but not the others.

Here we provide an example to illustrate the non-identifiability more intuitively. Setting $\mu_{00|1}(X) = \frac{1}{4}$, $\mu_{01|1}(X) = \frac{1}{4}$, $\mu_{10|1}(X) = \frac{1}{4}$, $\mu_{0|0}(X) = \frac{1}{4}$, and $\mu_{1|0}(X) = \frac{3}{4}$, we obtain two distinct solutions for the principal stratification scores:

Solution	$e_1(X)$	$e_2(X)$	$e_3(X)$	$e_4(X)$	$e_5(X)$	$e_6(X)$
Solution 1	$\frac{1}{4}$	0	$\frac{1}{4}$	$\frac{1}{4}$	0	0
Solution 2	$\frac{1}{4}$	$\frac{1}{4}$	0	$\frac{1}{4}$	0	$\frac{1}{4}$

This demonstrates the fundamental non-identifiability of principal stratification scores. While the conditional probabilities (and consequently $c(X)$) are estimable, the score $e_2(X)$ remains non-identifiable. Crucially, when $s \neq 0$, this non-identifiability propagates to the value loss $R_{e_2}(\pi, \psi)$, making causal effect estimation infeasible without additional assumptions. $\square$

Theorem 4.2 Let $\pi : X \rightarrow \{0, 1\}$ be a deterministic policy. For comparison policy $\psi \in \{\pi^0, \pi^1, \pi^o\}$, the worst-case expected value loss of π relative to ψ is given by:

$$R_{\text{sup}}(\pi, \psi) = C^\psi - \mathbb{E}[\pi(X)\{s \cdot b^\psi(X) + c(X)\}],$$

where C^ψ is a constant independent of π , and $b^\psi(X)$ is a key function depends on $\delta_l(\cdot)$, $\delta_u(\cdot)$ and $\mu(\cdot)$.

Proof of Theorem 4.2. Define $b(x) = s \cdot e_2(x) + c(x)$, and

$$\begin{aligned}L_b(x) &= \min_{e_2(x) \in [l(x), u(x)]} \{s \cdot e_2(x) + c(x)\}, \\ U_b(x) &= \max_{e_2(x) \in [l(x), u(x)]} \{s \cdot e_2(x) + c(x)\}.\end{aligned}$$

Note that the worst-case regret relative to the always and never treat policies are

$$\begin{aligned}R_{\text{sup}}(\pi, \pi^0) &= -E[\pi(X)L_b(X)], \\ R_{\text{sup}}(\pi, \pi^1) &= E[(1 - \pi(X))U_b(X)] = E[U_b(X)] - E[\pi(X)U_b(X)].\end{aligned}$$

From this, we can find the unconstrained minimax regret policies

$$\pi_0^* = \arg \min_{\pi} -E[\pi(X)L_b(X)] = \mathbb{I}\{L_b(X) \geq 0\},$$

$$\pi_1^* = \arg \min_{\pi} -E[\pi(X)U_b(X)] = \mathbb{I}\{U_b(X) \geq 0\}.$$

Now, the oracle policy is $\pi^o(x) = \mathbb{I}\{b(x) \geq 0\}$. So if $L_b(x) \geq 0 \iff \pi_0^*(x) = 1$, then $\pi^o(x) = 1$ for all possible values of the principal score $e_2(x)$. In this case,

$$\max_{e_2(x) \in [L(x), U(x)]} \{\pi^o(x) - \pi(x)\} b(x) = \{1 - \pi(x)\} U_b(x).$$

Similarly, if $U_b(x) < 0 \iff \pi^o(x) = 0$, and

$$\max_{e_2(x) \in [L(x), U(x)]} \{\pi^1(x) - \pi(x)\} b(x) = -\pi(x) L_b(x).$$

Finally, if $L_b(x) < 0$ and $U_b(x) \geq 0$ (so $\pi_0^*(x) = 0$ and $\pi_1^*(x) = 1$), then the oracle policy can be either 0 or 1, $\pi^o(x) \in \{0, 1\}$. Therefore,

$$\max_{e(x) \in [L(x), U(x)]} \{\pi^o(x) - \pi(x)\} b(x) = \max \{(1 - \pi(x)) U_b(x), -\pi(x) L_b(x)\} = U_b(x) - \pi(x) \{U_b(x) + L_b(x)\}.$$

Putting together the pieces, the worst-case regret relative to the oracle is

$$\begin{aligned} R_{\text{sup}}(\pi, \pi_o) &= \mathbb{E}([\pi_o^*(X) + (1 - \pi_o^*(X))\pi_1^*(X)]U_b(X)) \\ &\quad - \mathbb{E}[\pi(X)\{\pi_o^*(X)U_b(X) + (1 - \pi_1^*(X))L_b(X) \\ &\quad + (1 - \pi_o^*(X))\pi_1^*(X)(U_b(X) + L_b(X))\}]. \end{aligned}$$

To complete the proof, we need to compute $L_b(x)$ and $U_b(x)$.

For $s \geq 0$:

$$L_b(x) = s \cdot L(x) + c(x) = s \cdot \{(\delta_l(x)(\mu_{01|1}(x) - \mu_{1|0}(x)))\} + c(x),$$

$$U_b(x) = s \cdot U(x) + c(x) = s \cdot \{\mu_{01|1}(X) + \delta_u(x)(\mu_{11|1}(x) - \mu_{1|0}(x))\} + c(x).$$

For $s < 0$:

$$L_b(x) = s \cdot U(x) + c(x) = s \cdot \{\mu_{01|1}(X) + \delta_u(x)(\mu_{11|1}(x) - \mu_{1|0}(x))\} + c(x),$$

$$U_b(x) = s \cdot L(x) + c(x) = s \cdot \{(\delta_l(x)(\mu_{01|1}(x) - \mu_{1|0}(x)))\} + c(x).$$

Substituting the expressions for $L_b(x)$ and $U_b(x)$ into $R_{\text{sup}}(\pi, \psi)$ and simplifying, we obtain the following result.

For $s \geq 0$,

$$\begin{aligned} b^{\pi^o}(X) &= \delta_l(X)(\mu_{01|1}(X) - \mu_{1|0}(X)), \\ b^{\pi^1}(X) &= \mu_{01|1}(X) + \delta_u(X)(\mu_{11|1}(X) - \mu_{1|0}(X)), \\ b^{\pi^o}(X) &= \pi_0^*(X)\{\mu_{01|1}(X) + \delta_u(X)(\mu_{11|1}(X) - \mu_{1|0}(X))\} \\ &\quad + (1 - \pi_1^*(X))\{\delta_l(X)(\mu_{01|1}(X) - \mu_{1|0}(X))\} \\ &\quad + (1 - \pi_0^*(X))\pi_1^*(X)(\mu_{01|1}(X) + \delta_u(X)\{\mu_{11|1}(X) - \mu_{1|0}(X)\}) \\ &\quad + \delta_l(X)\{\mu_{01|1}(X) - \mu_{1|0}(X)\}. \end{aligned}$$

For $s < 0$,

$$\begin{aligned} b^{\pi^0}(X) &= \mu_{01|1}(X) + \delta_u(X)(\mu_{11|1}(X) - \mu_{1|0}(X)), \\ b^{\pi^1}(X) &= \delta_l(X)(\mu_{01|1}(X) - \mu_{1|0}(X)), \\ b^{\pi^o}(X) &= \pi_0^*(X)\{\mu_{01|1}(X) + \delta_l(X)(\mu_{01|1}(X) - \mu_{1|0}(X))\} \\ &\quad + (1 - \pi_1^*(X))\{\mu_{01|1}(X) + \delta_u(X)(\mu_{11|1}(X) - \mu_{1|0}(X))\} \\ &\quad + (1 - \pi_0^*(X))\pi_1^*(X)(\mu_{01|1}(X) + \delta_u(X)\{\mu_{11|1}(X) - \mu_{1|0}(X)\} \\ &\quad + \delta_l(X)\{\mu_{01|1}(X) - \mu_{1|0}(X)\}). \end{aligned}$$

□

E. Proofs in Section 5.1 (Additional Theorems in Appendix B.2)

Theorem B.1 (Bias and Double Robustness of the P-DR Estimator). *(a) Given the imputed outcome $\mu_{01|1}(x_i)$, and learned propensity score $\hat{e}(x_i) \in (0, 1)$ for all user-item pairs, the bias of the P-DR estimator is:*

$$\text{Bias}(\hat{\mu}_{01|1}^{P-DR}(x_i)) = \left| \frac{(e(x_i) - \hat{e}(x_i))\tilde{\mu}_{01|1}(x_i)}{\hat{e}(x_i)} \right|.$$

(b) The P-DR estimator is unbiased when either imputed outcome $\hat{\mu}_{01|1}(x_i)$ or learned propensities $\hat{e}(x_i) \in (0, 1)$ are accurate for all user-item pairs.

Proof of Theorem B.1. The P-DR estimator is given as

$$\hat{\mu}_{01|1}^{P-DR}(x_i) = \hat{\mu}_{01|1}(x_i) + \frac{T_i[(1 - I_i(1))Y_i(1) - \hat{\mu}_{01|1}(x_i)]}{\hat{e}(x_i)}.$$

By definition, the bias of the P-DR estimator is

$$\text{Bias}(\hat{\mu}_{01|1}^{P-DR}(x_i)) = |(1 - I_i(1))Y_i(1) - \mathbb{E}_T[\hat{\mu}_{01|1}^{P-DR}(x_i)]|.$$

The second term on the right hand side can be expanded as

$$\mathbb{E}_T[\hat{\mu}_{01|1}^{P-DR}(x_i)] = \hat{\mu}_{01|1}(x_i) + \frac{e(x_i)[(1 - I_i(1))Y_i(1) - \hat{\mu}_{01|1}(x_i)]}{\hat{e}(x_i)}.$$

By substituting the above formula into the bias of P-DR, we have

$$\begin{aligned} \text{Bias}(\hat{\mu}_{01|1}^{P-DR}(x_i)) &= \left| (1 - I_i(1))Y_i(1) - \hat{\mu}_{01|1}(x_i) - \frac{e(x_i)[(1 - I_i(1))Y_i(1) - \hat{\mu}_{01|1}(x_i)]}{\hat{e}(x_i)} \right| \\ &= \left| \frac{(e(x_i) - \hat{e}(x_i))[(1 - I_i(1))Y_i(1) - \hat{\mu}_{01|1}(x_i)]}{\hat{e}(x_i)} \right|, \end{aligned}$$

where the first term $e(x_i) - \hat{e}(x_i)$ on the right hind side equals to 0, when the learned propensities $\hat{e}(x_i)$ are accurate for all user-item pairs, i.e. $e(x_i) = \hat{e}(x_i)$. The second term on the right hind side equals to 0, when both of the outcome $(1 - I_i(1))Y_i(1) - \hat{\mu}_{01|1}(x_i)$ are accurate for all user-item pairs, i.e., $(1 - I_i(1))Y_i(1) = \hat{\mu}_{01|1}(x_i)$. □

Theorem B.2 (Variance of P-DR Estimator). *Given imputed outcomes $\hat{\mu}_{01|1}(x_i)$, and learned propensities $\hat{e}(x_i) \in (0, 1)$ for all user-item pairs, the variance of the P-DR estimator is*

$$\text{Var}(\hat{\mu}_{01|1}^{P-DR}(x_i)) = e(x_i)(1 - e(x_i)) \left[\frac{\tilde{\mu}_{01|1}(x_i)}{\hat{e}(x_i)} \right]^2.$$

Proof of Theorem B.2. The P-DR estimator is given as

$$\hat{\mu}_{01|1}^{P-DR}(x_i) = \hat{\mu}_{01|1}(x_i) + \frac{T_i[(1 - I_i(1))Y_i(1) - \hat{\mu}_{01|1}(x_i)]}{\hat{e}(x_i)}.$$

The variance of the P-DR estimator on the intervention assignment t_i is

$$\begin{aligned} \text{Var}(\hat{\mu}_{01|1}^{P-DR}(x_i)) &= \text{Var}(T_i) \left[\frac{(1 - I_i(1))Y_i(1) - \hat{\mu}_{01|1}(x_i)}{\hat{e}(x_i)} \right]^2 \\ &= e(x_i)(1 - e(x_i)) \left[\frac{(1 - I_i(1))Y_i(1) - \hat{\mu}_{01|1}(x_i)}{\hat{e}(x_i)} \right]^2. \end{aligned}$$

□

Lemma E.1 (Hoeffding's inequality). *Let $X_1, \dots, X_m$ be independent random variables with X_i taking values in $[a_i, r_i]$ for all $i \in [m]$. Then, for any $\epsilon > 0$, the following inequalities hold for $S_m = \sum_{i=1}^m X_i$:*

$$\mathbb{P}[S_m - \mathbb{E}[S_m] \geq \epsilon] \leq e^{-2\epsilon^2 / \sum_{i=1}^m (r_i - a_i)^2},$$

$$\mathbb{P}[S_m - \mathbb{E}[S_m] \leq -\epsilon] \leq e^{-2\epsilon^2 / \sum_{i=1}^m (r_i - a_i)^2}.$$

Proof. The proof can be found in Appendix D.1 of (Mohri et al., 2018).

Theorem B.3 (Tail Bound of P-DR Estimator). *Given imputed outcome $\hat{\mu}_{01|1}(x_i)$, and learned propensities $\hat{e}(x_i) \in (0, 1)$, with probability $1 - \eta$, the deviation of the P-DR estimator from its expectation has the following tail bound:*

$$\left| \hat{\mu}_{01|1}^{P-DR} - \mathbb{E}(\hat{\mu}_{01|1}^{P-DR}) \right| \leq \sqrt{\frac{\log\left(\frac{2}{\eta}\right)}{2n^2} \sum_{i=1}^n \left\{ \frac{\tilde{\mu}_{01|1}(x_i)}{\hat{e}(x_i)} \right\}^2}.$$

Proof of Theorem B.3. Since we assume that each intervention assignment T_i follows a Bernoulli distribution with probability $e(x_i)$, we can rewrite the random variable $\hat{\mu}_{01|1}^{P-DR}(x_i)$ as follows:

$$\begin{aligned} \mathbb{P}\left(\hat{\mu}_{01|1}^{P-DR}(x_i) = \hat{\mu}_{01|1}(x_i) + \frac{(1 - I_i(1))Y_i(1) - \hat{\mu}_{01|1}(x_i)}{\hat{e}(x_i)}\right) &= e(x_i), \\ \mathbb{P}\left(\hat{\mu}_{01|1}^{P-DR}(x_i) = \hat{\mu}_{01|1}(x_i)\right) &= 1 - e(x_i). \end{aligned}$$

Since $\hat{\mu}_{01|1}^{P-DR}(x_i)$ is a bounded random variable. The independence of $\{\hat{\mu}_{01|1}^{P-DR}(x_i)\}$ can be directly deduced from the independence of $\{T_i\}$. Therefore, according to Hoeffding's inequality in Lemma 1, for any $\epsilon > 0$, we have the following inequality

$$\begin{aligned} &\mathbb{P}\left(\left|\frac{1}{n} \sum_{i=1}^n \hat{\mu}_{01|1}^{P-DR}(x_i) - \mathbb{E}_T \left[\frac{1}{n} \sum_{i=1}^n \hat{\mu}_{01|1}^{P-DR}(x_i) \right]\right| \geq \epsilon\right) \\ &\leq 2 \exp\left(-\frac{2\epsilon^2 n^2}{\sum_{i=1}^n \{[(1 - I_i(1))Y_i(1) - \hat{\mu}_{01|1}(x_i)]/\hat{e}(x_i)\}^2}\right). \end{aligned}$$

Setting the right-hand side of the inequality to η and solving for ϵ completes the proof.

□

F. Proofs in Section 5.3

F.1. Notation

First, we let n represents the total sample size.

Second, we let $\delta(\cdot)$ refers to $\delta_l(\cdot)$ and $\delta_u(\cdot)$, and $\hat{\delta}(\cdot)$ refers to $\hat{\delta}_l(\cdot)$ and $\hat{\delta}_u(\cdot)$. Similarly, we let $\mu(\cdot)$ refers joint probabilities $\mu_{00|1}(\cdot)$, $\mu_{01|1}(\cdot)$ and so on, while $\hat{\mu}(\cdot)$ refers to the P-DR estimators of them. Next, we let $b^\psi(X_i; \delta(\cdot), \mu(\cdot))$ represents $b^\psi(X_i)$ depending on $\delta(\cdot)$ and $\mu(\cdot)$, e.g. $b^{\pi^0}(X_i; \hat{\delta}(\cdot), \hat{\mu}(\cdot)) = \hat{\delta}_l(X_i)(\mu_{01|1}^{P-DR}(X_i) - \mu_{1|0}^{P-DR}(X_i))$ (when $s \geq 0$), and the notation $c(X_i; \mu(\cdot))$ has a similar meaning.

Finally, we let

$$\tilde{R}_{\text{sup}}(\pi, \delta(\cdot), \mu(\cdot); \psi) = C^\psi - \frac{1}{n} \sum_{i=1}^n \pi(X_i) \{s \cdot b^\psi(X_i; \delta(\cdot), \mu(\cdot)) + c(X_i; \mu(\cdot))\}.$$

With these quantities, we can proceed with the proof of the theorem in Section 5.3.

F.2. Proof of Theorem 5.1 and Theorem 5.2

Theorem 5.4. *Let $\hat{\pi}$ solve Eqn (1) with alternative policy $\psi = \pi^1$ (if $s < 0$) or $\psi = \pi^0$ (if $s \geq 0$) and no constraint, the estimates $\hat{e}$, $\hat{\mu}^{P-DR}$ and $\hat{\delta}_l$ fitted on a separate sample. The excess worst-case expected value loss of $\hat{\pi}$ relative to π_ψ^* satisfies*

$$\begin{aligned} R_{\text{sup}}(\hat{\pi}, \psi) - R_{\text{sup}}(\pi_\psi^*, \psi) &\leq U \cdot \left\{ \frac{12 + \eta}{\eta} (2R_n(\Pi) + \frac{t}{\sqrt{n}}) + \|\hat{\gamma}_0 - \gamma_0\|_2 \|\hat{\mu}_{1|0} - \mu_{1|0}\|_2 \right. \\ &\quad \left. + \|\hat{\gamma}_1 - \gamma_1\|_2 (\|\hat{\mu}_{00|1} - \mu_{00|1}\|_2 + \|\hat{\mu}_{01|1} - \mu_{01|1}\|_2 + \|\hat{\mu}_{10|1} - \mu_{10|1}\|_2) \right\} \\ &\quad + 2|s| \cdot (R_l(\hat{\delta}_l) + \frac{t}{2\sqrt{n}}), \end{aligned}$$

with a probability of at least $1 - 2 \exp\left(-\frac{t^2}{2}\right)$, where η is the overlap parameter in Assumption 3.1, U is a constant depending on the return values, $\|\hat{\gamma}_1 - \gamma_1\|_2^2 \equiv \mathbb{E} \left[\left\{ \frac{T}{e(X)} - \frac{T}{\hat{e}(X)} \right\}^2 \right]$, $\|\hat{\gamma}_0 - \gamma_0\|_2^2 \equiv \mathbb{E} \left[\left\{ \frac{1-T}{1-e(X)} - \frac{1-T}{1-\hat{e}(X)} \right\}^2 \right]$, and $\|\hat{\mu}_{01|1}(\cdot) - \mu_{01|1}(\cdot)\|_2^2 \equiv \mathbb{E} \left[\{\hat{\mu}_{01|1}(X) - \mu_{01|1}(X)\}^2 \right]$.

Proof of Theorem 5.1. This follows directly from combining Lemmas F.1 and F.2 below via the union bound.

Theorem 5.5. *Let $\hat{\pi}_o$ solve Eqn 1 with the alternative policy $\psi = \pi^o$ and no constraint, the estimates $\hat{e}$, $\hat{\mu}^{P-DR}$, $\hat{\delta}_l$, $\hat{\delta}_u$ and estimated policies $\hat{\pi}_0$, $\hat{\pi}_1$ fitted on a separate sample. The excess worst-case expected value loss of $\hat{\pi}_o$ relative to π_o^* satisfies:*

$$\begin{aligned} R_{\text{sup}}(\hat{\pi}_o, \psi) - R_{\text{sup}}(\pi_o^*, \psi) &\leq U \cdot \left\{ \frac{12 + \eta}{\eta} (2R_n(\Pi) + \frac{t}{\sqrt{n}}) + \|\hat{\gamma}_0 - \gamma_0\|_2 \|\hat{\mu}_{1|0} - \mu_{1|0}\|_2 \right. \\ &\quad \left. + \|\hat{\gamma}_1 - \gamma_1\|_2 (\|\hat{\mu}_{00|1} - \mu_{00|1}\|_2 + \|\hat{\mu}_{01|1} - \mu_{01|1}\|_2 + \|\hat{\mu}_{10|1} - \mu_{10|1}\|_2) \right\} \\ &\quad + 4 \sum_{w=0}^1 \{R_{\text{sup}}(\hat{\pi}_w, \pi_w) - R_{\text{sup}}(\pi_w^*, \pi_w)\} + 2s \cdot (R_l(\hat{\delta}_l) + R_u(\hat{\delta}_u) + \frac{t}{2\sqrt{n}}), \end{aligned}$$

with a probability of at least $1 - 2 \exp\left(-\frac{t^2}{2}\right)$, where U is a constant depending on the return values.

Proof of Theorem 5.2. This follows directly from combining Lemmas F.1 and F.3 below via the union bound.

Lemma F.1. *$\hat{\pi}$ is the optimal policy with alternative policy $\psi = \pi^1$ (if $s < 0$) or $\psi = \pi^0$ (if $s \geq 0$), and with estimates $\hat{e}$ and $\hat{\mu}^{P-DR}$ and classifier $\hat{\delta}_l$ fit on a separate sample. The excess worst-case expected value loss of $\hat{\pi}$ relative to π_ψ^* satisfies*

$$\begin{aligned} R_{\text{sup}}(\hat{\pi}, \psi) - R_{\text{sup}}(\pi_\psi^*, \psi) &\leq U \times \left\{ \frac{12 + \eta}{\eta} \times (2R_n(\Pi) + \frac{t}{\sqrt{n}}) + \|\hat{\gamma}_0 - \gamma_0\|_2 \|\hat{\mu}_{1|0} - \mu_{1|0}\|_2 \right. \\ &\quad \left. + \|\hat{\gamma}_1 - \gamma_1\|_2 (\|\hat{\mu}_{00|1} - \mu_{00|1}\|_2 + \|\hat{\mu}_{01|1} - \mu_{01|1}\|_2 + \|\hat{\mu}_{10|1} - \mu_{10|1}\|_2) \right\} \\ &\quad + 2 \sup_{\pi \in \Pi} \left| \tilde{R}_{\text{sup}}(\pi, \hat{\delta}(\cdot), \mu(\cdot); \psi) - \tilde{R}_{\text{sup}}(\pi, \delta(\cdot), \mu(\cdot); \psi) \right|, \end{aligned}$$

with probability at least $1 - 2 \exp\left(-\frac{t^2}{2}\right)$, where $n = n$ is the sample size, η is the overlap parameter in Assumption 1, U is a constant depending on the return values, $\|\hat{\gamma}_1 - \gamma_1\|_2^2 \equiv \mathbb{E}\left[\left\{\frac{T}{e(X)} - \frac{T}{\hat{e}(X)}\right\}^2\right]$ and $\|\hat{\mu}_{01|1}(\cdot) - \mu_{01|1}(\cdot)\|_2^2 \equiv \mathbb{E}\left[\{\hat{\mu}_{01|1}(X) - \mu_{01|1}(X)\}^2\right]$.

Proof of Lemma F.1. First, note that the excess regret can be decomposed into

$$\begin{aligned} R_{\text{sup}}(\hat{\pi}, \psi) - R_{\text{sup}}(\pi^*, \psi) &= R_{\text{sup}}(\hat{\pi}, \psi) - \hat{R}_{\text{sup}}(\hat{\pi}, \psi) + \hat{R}_{\text{sup}}(\hat{\pi}, \psi) - \hat{R}_{\text{sup}}(\pi^*, \psi) + \hat{R}_{\text{sup}}(\pi^*, \psi) - R_{\text{sup}}(\pi^*, \psi) \\ &\leq |R_{\text{sup}}(\hat{\pi}, \psi) - \hat{R}_{\text{sup}}(\hat{\pi}, \psi)| + |\hat{R}_{\text{sup}}(\pi^*, \psi) - R_{\text{sup}}(\pi^*, \psi)| \\ &\leq 2 \sup_{\pi \in \Pi} \left| \hat{R}_{\text{sup}}(\pi, \psi) - R_{\text{sup}}(\pi, \psi) \right|, \end{aligned}$$

where we have used that $\hat{\pi}$ minimizes $\hat{R}_{\text{sup}}(\pi, \psi)$, i.e., $\hat{R}_{\text{sup}}(\hat{\pi}, \psi) - \hat{R}_{\text{sup}}(\pi^*, \psi) \leq 0$.

We further decompose $\hat{R}_{\text{sup}}(\pi, \psi) - R_{\text{sup}}(\pi, \psi)$ into

$$\begin{aligned} \hat{R}_{\text{sup}}(\pi, \psi) - R_{\text{sup}}(\pi, \psi) &= \hat{R}_{\text{sup}}(\pi, \psi) - \tilde{R}_{\text{sup}}(\pi, \hat{\delta}(\cdot), \mu(\cdot); \psi) \quad (a) \\ &\quad + \tilde{R}_{\text{sup}}(\pi, \delta(\cdot), \mu(\cdot); \psi) - R_{\text{sup}}(\pi, \psi) \quad (b) \\ &\quad + \tilde{R}_{\text{sup}}(\pi, \hat{\delta}(\cdot), \mu(\cdot); \psi) - \tilde{R}_{\text{sup}}(\pi, \delta(\cdot), \mu(\cdot); \psi). \end{aligned}$$

We will now control terms (a) and (b), following closely the proof of Lemma 4 in (Athey & Wager, 2021). For the P-DR estimator, we have following decompositions (here we only show $\mu_{01|1}^{P-DR}(x)$ and $\mu_{1|0}^{P-DR}(x)$):

$$\begin{aligned} \mu_{01|1}^{P-DR}(x) - \mu_{01|1}(x) &= \hat{\mu}_{01|1}(x) - \mu_{01|1}(x) + \frac{T}{\hat{e}(x)} \{(1-C)Y - \hat{\mu}_{01|1}(x)\} \\ &= \{\hat{\mu}_{01|1}(x) - \mu_{01|1}(x)\} \left\{1 - \frac{T}{e(x)}\right\} + \frac{T}{\hat{e}(x)} \{(1-C)Y - \mu_{01|1}(x)\} \\ &\quad + \left\{\frac{T}{\hat{e}(x)} - \frac{T}{e(x)}\right\} \{\mu_{01|1}(x) - \hat{\mu}_{01|1}(x)\}, \end{aligned}$$

$$\begin{aligned} \mu_{1|0}^{P-DR}(x) - \mu_{1|0}(x) &= \hat{\mu}_{1|0}(x) - \mu_{1|0}(x) + \frac{1-T}{1-\hat{e}(x)} \{Y - \hat{\mu}_{1|0}(x)\} \\ &= \{\hat{\mu}_{1|0}(x) - \mu_{1|0}(x)\} \left\{1 - \frac{1-T}{1-e(x)}\right\} + \frac{1-T}{1-\hat{e}(x)} \{Y - \mu_{1|0}(x)\} \\ &\quad + \left\{\frac{1-T}{1-\hat{e}(x)} - \frac{1-T}{1-e(x)}\right\} \{\mu_{1|0}(x) - \hat{\mu}_{1|0}(x)\}, \end{aligned}$$

then we rewrite $c(x)$ as:

$$c(x) = \mu_{00|1}(x)(r_1 - r_5) + \mu_{01|1}(x)(r_3 - r_6) + \mu_{10|1}(x)(r_4 - r_5) + \mu_{1|0}(x)(-r_5 + r_6) + r_5.$$

With these, we can compute the expectation of term (a):

$$\begin{aligned} \mathbb{E}[(a)] &= \mathbb{E}[\pi(x) \{s \cdot \{b^\psi(x; \hat{\delta}(\cdot), \mu(\cdot)) - b^\psi(x; \hat{\delta}(\cdot), \hat{\mu}(\cdot))\} + c(x; \hat{\mu}(\cdot)) - c(x; \mu(\cdot))\}] \\ &= \mathbb{E}[\pi(x) \{s \cdot \hat{\delta}_l(x) \cdot \{\mu_{01|1}(x) - \mu_{1|0}(x) - \mu_{01|1}^{P-DR}(x) + \mu_{1|0}^{P-DR}(x)\} + c(x; \hat{\mu}(\cdot)) - c(x; \mu(\cdot))\}] \\ &= \mathbb{E}[\pi(x) \{(r_1 - r_5)(\mu_{00|1}(x) - \mu_{00|1}^{P-DR}(x)) + (s \cdot \hat{\delta}_l(x) + r_3 - r_6)(\mu_{01|1}(x) - \mu_{01|1}^{P-DR}(x)) \\ &\quad + (r_4 - r_5)(\mu_{10|1}(x) - \mu_{10|1}^{P-DR}(x)) + (-s \cdot \hat{\delta}_l(x) - r_5 + r_6)(\mu_{1|0}(x) - \mu_{1|0}^{P-DR}(x))\}]. \end{aligned}$$

We have the fact that:

$$\mathbb{E}\left[\pi(x) \left\{1 - \frac{T}{e(x)}\right\}\right] = 0, \quad \mathbb{E}\left[\pi(x) \left\{1 - \frac{1-T}{1-e(x)}\right\}\right] = 0,$$

$$\begin{aligned}\mathbb{E}[\pi(x)\{(1-C)(1-Y) - \mu_{00|1}(x)\}] &= 0, & \mathbb{E}[\pi(x)\{(1-C)Y - \mu_{01|1}(x)\}] &= 0, \\ \mathbb{E}[\pi(x)\{C(1-Y) - \mu_{10|1}(x)\}] &= 0, & \mathbb{E}[\pi(x)\{Y - \mu_{1|0}(x)\}] &= 0,\end{aligned}$$

and $\hat{\delta}_l, \hat{\mu}$ and $\hat{e}$ come from a different sample. Hence, We can transform the form of $\mathbb{E}[(a)]$ into:

$$\begin{aligned}\mathbb{E}[(a)] &= \mathbb{E} \left[\pi(x)(r_1 - r_5) \left\{ \frac{T}{e(x)} - \frac{T}{\hat{e}(x)} \right\} \{ \mu_{00|1}(x) - \hat{\mu}_{00|1}(x) \} \right] \\ &+ \mathbb{E} \left[\pi(x)\{s \cdot \hat{\delta}_l(x) + r_3 - r_6\} \left\{ \frac{T}{e(x)} - \frac{T}{\hat{e}(x)} \right\} \{ \mu_{01|1}(x) - \hat{\mu}_{01|1}(x) \} \right] \\ &+ \mathbb{E} \left[\pi(x)(r_4 - r_5) \left\{ \frac{T}{e(x)} - \frac{T}{\hat{e}(x)} \right\} \{ \mu_{10|1}(x) - \hat{\mu}_{10|1}(x) \} \right] \\ &+ \mathbb{E} \left[\pi(x)\{-s \cdot \hat{\delta}_l(x) - r_5 + r_6\} \left\{ \frac{1-T}{1-e(x)} - \frac{1-T}{1-\hat{e}(x)} \right\} \{ \mu_{1|0}(x) - \hat{\mu}_{1|0}(x) \} \right].\end{aligned}$$

And:

$$\mathbb{E}[(b)] = C^{r_\psi} - \mathbb{E}[(\pi(x))\{s \cdot b^\psi(x) + c(x)\}] - R_{sup}(\pi, \psi) = 0.$$

Now we define a function $f : \mathcal{X} \rightarrow \mathbb{R}$ as:

$$\begin{aligned}f_\pi(x, t, c, y; \psi) &= \pi(x)[(r_1 - r_5)(\mu_{00|1}(x) - \mu_{00|1}^{P-DR}(x)) + (s \cdot \hat{\delta}_l(x) + r_3 - r_6)(\mu_{01|1}(x) - \mu_{01|1}^{P-DR}(x)) \\ &+ (r_4 - r_5)(\mu_{10|1}(x) - \mu_{10|1}^{P-DR}(x)) + (-s \cdot \hat{\delta}_l(x) - r_5 + r_6)(\mu_{1|0}(x) - \mu_{1|0}^{P-DR}(x))] \\ &- \pi(x)\{s \cdot b^\psi(x) + c(x)\},\end{aligned}$$

and the function class $\mathcal{F} \equiv \{f_\pi | \pi \in \Pi\}$ as the set of all functions f as we vary π in Π .

Then we can rewrite (a),(b) as

$$(a) + (b) = \frac{1}{n} \sum_{i=1}^n f_\pi(X_i, T_i, I_i, Y_i; \psi) - R_{sup}(\pi, \psi)$$

,

and

$$\mathbb{E}[f_\pi(x, t, c, y; \psi)] = R_{sup} + \mathbb{E}[(a)].$$

Putting these together, we have:

$$\begin{aligned}|(a) + (b)| &= \left| \frac{1}{n} \sum_{i=1}^n f_\pi(X_i, T_i, I_i, Y_i; \psi) - R_{sup}(\pi, \psi) \right| \\ &= \left| \frac{1}{n} \sum_{i=1}^n f_\pi(X_i, T_i, I_i, Y_i; \psi) - \mathbb{E}[f_\pi(x, t, c, y; \psi)] \right| + \left| \mathbb{E}[f_\pi(x, t, c, y; \psi)] - R_{sup}(\pi, \psi) \right| \\ &\leq \left| \frac{1}{n} \sum_{i=1}^n f_\pi(X_i, T_i, I_i, Y_i; \psi) - \mathbb{E}[f_\pi(x, t, c, y; \psi)] \right| \\ &+ \left| \mathbb{E} \left[\pi(x)(r_1 - r_5) \left\{ \frac{T}{e(x)} - \frac{T}{\hat{e}(x)} \right\} \{ \mu_{00|1}(x) - \hat{\mu}_{00|1}(x) \} \right] \right| \\ &+ \left| \mathbb{E} \left[\pi(x)\{s \cdot \hat{\delta}_l(x) + r_3 - r_6\} \left\{ \frac{T}{e(x)} - \frac{T}{\hat{e}(x)} \right\} \{ \mu_{01|1}(x) - \hat{\mu}_{01|1}(x) \} \right] \right| \\ &+ \left| \mathbb{E} \left[\pi(x)(r_4 - r_5) \left\{ \frac{T}{e(x)} - \frac{T}{\hat{e}(x)} \right\} \{ \mu_{10|1}(x) - \hat{\mu}_{10|1}(x) \} \right] \right| \\ &+ \left| \mathbb{E} \left[\pi(x)\{-s \cdot \hat{\delta}_l(x) - r_5 + r_6\} \left\{ \frac{1-T}{1-e(x)} - \frac{1-T}{1-\hat{e}(x)} \right\} \{ \mu_{1|0}(x) - \hat{\mu}_{1|0}(x) \} \right] \right|.\end{aligned}$$

With Assumption 3.1, we can get the upper bound of P-DR estimator's error

$$\begin{aligned}
 |\mu_{10|1}(x) - \mu_{10|1}^{P-DR}(x)| &\leq \left| \{\mu_{01|1}(x) - \hat{\mu}_{01|1}(x)\} \left\{1 - \frac{T}{e(x)}\right\} \right| + \left| \frac{T}{\hat{e}(x)} \{(1-C)Y - \mu_{01|1}(x)\} \right| \\
 &\quad + \left| \left\{ \frac{T}{\hat{e}(x)} - \frac{T}{e(x)} \right\} \{\mu_{01|1}(x) - \hat{\mu}_{01|1}(x)\} \right| \\
 &\leq \frac{1-\eta}{\eta} \|\hat{\mu} - \mu\|_{\infty} + \frac{1}{\eta} + \left\| \frac{1}{e} - \frac{1}{\hat{e}} \right\|_{\infty} \|\hat{\mu} - \mu\|_{\infty} \\
 &\leq \frac{1-\eta}{\eta} + \frac{1}{\eta} + \frac{1}{\eta} - \frac{1}{1-\eta} \\
 &\leq \frac{3}{\eta}.
 \end{aligned}$$

Similarly we can obtain that the upper bound of the estimation error for other joint probability estimators is also $\frac{3}{\eta}$.

Now notice that for $\psi \in \{\pi^0, \pi^1, \pi^o\}$, $|s \cdot b^{\psi}(x) + c(x)|$, $\hat{\delta}_l(x)$ and r_i are bounded by some constant U . Then for any (x, c, t, y) , we have

$$|f_{\pi}(x, t, c, y; \psi)| \leq U \cdot \frac{12 + \eta}{\eta}.$$

This also shows that the Rademacher complexity of $\mathcal{F}$ is

$$R_n(\mathcal{F}) = 2U \cdot \frac{12 + \eta}{\eta} \cdot R_n(\Pi).$$

So by Theorem 4.2 of (Wainwright, 2019), for any $n \geq 1$ and $t \geq 0$

$$\sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n f_{\pi}(X_i, T_i, I_i, Y_i; \psi) - \mathbb{E}[f_{\pi}(x, t, c, y; \psi)] \right| \leq 2U \cdot \frac{12 + \eta}{\eta} \cdot \left(2R_n(\Pi) + \frac{t}{\sqrt{n}} \right),$$

with a probability of at least $1 - \exp(-\frac{t^2}{2})$.

Finally, notice that by the Cauchy-Schwarz inequality,

$$\begin{aligned}
 &\left| \mathbb{E} \left[\pi(x) \{s \cdot \hat{\delta}_l(x) + r_3 - r_6\} \left\{ \frac{T}{e(x)} - \frac{T}{\hat{e}(x)} \right\} \{\mu_{01|1}(x) - \hat{\mu}_{01|1}(x)\} \right] \right| \\
 &\leq U \sqrt{\mathbb{E} \left[\left\{ \frac{T}{e(x)} - \frac{T}{\hat{e}(x)} \right\}^2 \right] \mathbb{E} [\{\mu_{01|1}(x) - \hat{\mu}_{01|1}(x)\}^2]}.
 \end{aligned}$$

Other terms can also be obtained in a similar manner.

Combining these two bounds, we can get the result of $|(a) + (b)|$ then the result of the lemma. $\square$

Lemma F.2. For $s \geq 0, \psi = \pi^0$ or $s < 0, \psi = \pi^1$,

$$\sup_{\pi \in \Pi} \left| \tilde{R}_{\text{sup}}(\pi, \hat{\delta}(\cdot), \mu(\cdot); \psi) - \tilde{R}_{\text{sup}}(\pi, \delta(\cdot), \mu(\cdot); \psi) \right| \leq |s| \cdot \left\{ R_l(\hat{\delta}_l) + \frac{t}{2\sqrt{n}} \right\},$$

with a probability of at least $1 - \exp(-\frac{t^2}{2})$.

Proof of Lemma F.2 .

$$\begin{aligned}
 \left| \tilde{R}_{\text{sup}}(\pi, \hat{\delta}(\cdot), \mu(\cdot); \psi) - \tilde{R}_{\text{sup}}(\pi, \delta(\cdot), \mu(\cdot); \psi) \right| &= \left| \frac{s}{n} \sum_{i=1}^n \pi(X_i) \{\hat{\delta}_l(X_i) - \delta_l(X_i)\} \{\mu_{01|1}(X_i) - \mu_{1|0}(X_i)\} \right| \\
 &\leq \frac{|s|}{n} \sum_{i=1}^n \mathbb{I}\{\hat{\delta}_l(X_i) \neq \delta_l(X_i)\} |\mu_{01|1}(X_i) - \mu_{1|0}(X_i)|.
 \end{aligned}$$

Note that

$$\mathbb{E} \left[\mathbb{I}\{\hat{\delta}_l(X) \neq \delta_l(X)\} \cdot |\mu_{01|1}(X) - \mu_{1|0}(X)| \right] = R_l(\hat{\delta}_l).$$

We know that each $\mathbb{I}\{\hat{\delta}_l(X_i) \neq \delta_l(X_i)\} |\mu_{01|1}(X_i) - \mu_{1|0}(X_i)|$ is bounded between 0 and 1, it is sub-Gaussian with scale parameter 1. Furthermore, they are independent across $i = 1, \dots, n$, so by the Hoeffding bound,

$$\Pr \left(\frac{s}{n} \sum_{i=1}^n \mathbb{I}\{\hat{\delta}_l(X_i) \neq \delta_l(X_i)\} |\mu_{01|1}(X_i) - \mu_{1|0}(X_i)| \leq |s| \cdot \left\{ R_l(\hat{\delta}_l) + \frac{t}{2\sqrt{n}} \right\} \right) \geq 1 - \exp\left(-\frac{t^2}{2}\right).$$

□

Lemma F.3.

$$\begin{aligned} & \sup_{\pi \in \Pi} \left| \tilde{R}_{\text{sup}}(\pi, \hat{\delta}(\cdot), \mu(\cdot); \pi^o) - \tilde{R}_{\text{sup}}(\pi, \delta(\cdot), \mu(\cdot); \pi^o) \right| \\ & \leq 2 \times \{R_{\text{sup}}(\hat{\pi}, \pi^1) - R_{\text{sup}}(\pi^*, \pi^1)\} + 2 \times \{R_{\text{sup}}(\hat{\pi}, \pi^o) - R_{\text{sup}}(\pi^*, \pi^o)\} \\ & + |s| \times \left(R_+(\hat{\delta}_+) + R_\tau(\hat{\delta}_\tau) + \frac{t}{2\sqrt{n}} \right), \end{aligned}$$

with a probability of at least $1 - \exp(-\frac{t^2}{2})$.

Proof of Lemma F.3. We only provide the proof for the case where $s \geq 0$ here, and the proof for $s < 0$ is similar. Define

$$\begin{aligned} \check{L}_b(x) &= s \cdot \{(\hat{\delta}_l(x)(\mu_{01|1}(x) - \mu_{1|0}(x)))\} + c(x), \\ \check{U}_b(x) &= s \cdot \{\mu_{01|1}(x) + \hat{\delta}_u(x)(\mu_{11|1}(x) - \mu_{1|0}(x))\} + c(x), \\ Q(x) &= \pi_0^*(x)U_b(x) + (1 - \pi_1^*(x))L_b(x) + (1 - \pi_0^*(x))\pi_1^*(x)(U_b(x) + L_b(x)), \\ \tilde{Q}(x) &= \hat{\pi}_0(x)U_b(x) + (1 - \hat{\pi}_1(x))L_b(x) + (1 - \hat{\pi}_0(x))\hat{\pi}_1(x)(U_b(x) + L_b(x)), \\ \check{Q}(x) &= \hat{\pi}_0(x)\check{U}_b(x) + (1 - \hat{\pi}_1(x))\check{L}_b(x) + (1 - \hat{\pi}_0(x))\hat{\pi}_1(x)(\check{U}_b(x) + \check{L}_b(x)). \end{aligned}$$

With these definitions, we can write

$$\begin{aligned} \left| \tilde{R}_{\text{sup}}(\pi, \hat{\delta}(\cdot), \mu(\cdot); \pi^o) - \tilde{R}_{\text{sup}}(\pi, \delta(\cdot), \mu(\cdot); \pi^o) \right| &= \left| \frac{1}{n} \sum_{i=1}^n \pi(X) \{Q(X_i) - \check{Q}(X_i)\} \right| \\ &= \left| \frac{1}{n} \sum_{i=1}^n \pi(X) \{Q(X_i) - \tilde{Q}(X_i)\} + \frac{1}{n} \sum_{i=1}^n \pi(X) \{\tilde{Q}(X_i) - \check{Q}(X_i)\} \right| \\ &\leq \frac{1}{n} \sum_{i=1}^n |Q(X_i) - \tilde{Q}(X_i)| + \frac{1}{n} \sum_{i=1}^n |\tilde{Q}(X_i) - \check{Q}(X_i)|. \end{aligned}$$

Working with the first term:

$$\begin{aligned} \tilde{Q}(x) - Q(x) &= (\hat{\pi}_1(x) - \pi_1^*(x))U_b(x) - (\hat{\pi}_1(x)\hat{\pi}_0(x) - \pi_1^*(x)\pi_0^*(x))(U_b(x) + L_b(x)) + (\hat{\pi}_0(x) - \pi_0^*(x))U_b(x) \\ &= (\hat{\pi}_0(x) - \pi_0^*(x)) \times (-L_b(x)\pi_1^*(x) + (1 - \pi_1^*(x))U_b(x)) \quad (*) \\ &+ (\hat{\pi}_1(x) - \pi_1^*(x)) \times (-L_b(x)\hat{\pi}_0(x) + (1 - \hat{\pi}_0^*(x))U_b(x)) \quad (**). \end{aligned}$$

Notice that $\pi_1^*(x) = 0 \Leftrightarrow U_b(x) \leq 0$, since $L_b(x) \leq U_b(x)$, this implies that when $\pi_1^*(x) = 0$, $|U_b(x)| \leq |L_b(x)|$. Therefore,

$$|*| \leq \mathbb{I}\{\hat{\pi}_0(x) \neq \pi_0^*(x)\} |L_b(x)|.$$

Similarly, if $\pi_0^*(x) = 1$, then $0 \leq L_b(x) \leq U_b(x)$, so $|L_b(x)| \leq |U_b(x)|$. Hence we have

$$\begin{aligned}
 |**| &\leq \mathbb{I}\{\hat{\pi}_1(x) \neq \pi_1^*(x)\} \mathbb{I}\{\hat{\pi}_0(x) = \pi_0^*(x)\} |L_b(x)\pi_0^*(x) + (1 - \pi_0^*(x))U_b(x)| \\
 &\quad + \mathbb{I}\{\hat{\pi}_0(x) \neq \pi_0^*(x)\} \mathbb{I}\{\hat{\pi}_1(x) \neq \pi_1^*(x)\} |\hat{\pi}_0(x)L_b(x) + (1 - \hat{\pi}_0(x))U_b(x)| \\
 &\leq \mathbb{I}\{\hat{\pi}_1(x) \neq \pi_1^*(x)\} \mathbb{I}\{\hat{\pi}_0(x) = \pi_0^*(x)\} |U_b(x)| \\
 &\quad + \mathbb{I}\{\hat{\pi}_0(x) \neq \pi_0^*(x)\} \mathbb{I}\{\hat{\pi}_1(x) \neq \pi_1^*(x)\} |L_b(x)| + \mathbb{I}\{\hat{\pi}_0(x) \neq \pi_0^*(x)\} \mathbb{I}\{\hat{\pi}_1(x) \neq \pi_1^*(x)\} |U_b(x)| \\
 &\leq \mathbb{I}\{\hat{\pi}_1(x) \neq \pi_1^*(x)\} |U_b(x)| + \mathbb{I}\{\hat{\pi}_0(x) \neq \pi_0^*(x)\} |L_b(x)| + \mathbb{I}\{\hat{\pi}_1(x) \neq \pi_1^*(x)\} |U_b(x)| \\
 &= 2\mathbb{I}\{\hat{\pi}_1(x) \neq \pi_1^*(x)\} |U_b(x)| + \mathbb{I}\{\hat{\pi}_0(x) \neq \pi_0^*(x)\} |L_b(x)|.
 \end{aligned}$$

Putting together these pieces, we have

$$|Q(x) - \tilde{Q}(x)| \leq 2\mathbb{I}\{\hat{\pi}_0(x) \neq \pi_0^*(x)\} |L_b(x)| + 2\mathbb{I}\{\hat{\pi}_1(x) \neq \pi_1^*(x)\} |U_b(x)|.$$

So the expectation is bounded by the regrets based on Theorem 5.1

$$\begin{aligned}
 \mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n |Q(X_i) - \tilde{Q}(X_i)| \right] &\leq 2\mathbb{E} [\mathbb{I}\{\hat{\pi}_0(x) \neq \pi_0^*(x)\} |L_b(x)|] + 2\mathbb{E} [\mathbb{I}\{\hat{\pi}_1(x) \neq \pi_1^*(x)\} |U_b(x)|] \\
 &= 2 \times \{R_{\text{sup}}(\hat{\pi}_1, \pi_1^*) - R_{\text{sup}}(\pi_1^*, \pi_1^*)\} + 2 \times \{R_{\text{sup}}(\hat{\pi}_0, \pi_0^*) - R_{\text{sup}}(\pi_0^*, \pi_0^*)\}.
 \end{aligned}$$

We know $|Q(X_i) - \hat{Q}(X_i)|$ can be bounded by U , so by the Hoeffding inequation:

$$\begin{aligned}
 \Pr \left(\frac{1}{n} \sum_{i=1}^n |Q(X_i) - \tilde{Q}(X_i)| \leq 2 \{R_{\text{sup}}(\hat{\pi}_1, \pi_1^*) - R_{\text{sup}}(\pi_1^*, \pi_1^*)\} + 2 \{R_{\text{sup}}(\hat{\pi}_0, \pi_0^*) - R_{\text{sup}}(\pi_0^*, \pi_0^*)\} + \frac{t}{2\sqrt{n}} \right) \\
 \geq 1 - \exp\left(-\frac{t^2}{2}\right).
 \end{aligned}$$

Then we come to discuss the second term:

$$\begin{aligned}
 |\tilde{Q}(x) - \check{Q}(x)| &= |(L_b(x) - \check{L}_b(x))(1 - \hat{\pi}_1 + (1 - \hat{\pi}_0)\hat{\pi}_1) + (U_b(x) - \check{U}_b(x))(\hat{\pi}_0 + (1 - \hat{\pi}_0)\hat{\pi}_1)| \\
 &\leq |L_b(x) - \check{L}_b(x)| + |U_b(x) - \check{U}_b(x)|.
 \end{aligned}$$

To re-write this, notice that

$$\begin{aligned}
 |L_b(x) - \check{L}_b(x)| &= |s| \cdot \mathbb{I}\{\hat{\delta}_l(x) \neq \delta_l(x)\} |\mu_{01|1}(x) - \mu_{1|0}(x)|, \\
 |U_b(x) - \check{U}_b(x)| &= |s| \cdot \mathbb{I}\{\hat{\delta}_u(x) \neq \delta_u(x)\} |\mu_{11|1}(x) - \mu_{1|0}(x)|.
 \end{aligned}$$

Hence

$$\frac{|\tilde{Q}(x) - \check{Q}(x)|}{s} \leq \mathbb{I}\{\hat{\delta}_l(x) \neq \delta_l(x)\} |\mu_{01|1}(x) - \mu_{1|0}(x)| + \mathbb{I}\{\hat{\delta}_u(x) \neq \delta_u(x)\} |\mu_{11|1}(x) - \mu_{1|0}(x)|,$$

$$\begin{aligned}
 \mathbb{E} \left[\frac{1}{s} \frac{1}{n} \sum_{i=1}^n |\tilde{Q}(X_i) - \check{Q}(X_i)| \right] &\leq \mathbb{E} [\mathbb{I}\{\hat{\delta}_l(x) \neq \delta_l(x)\} |\mu_{01|1}(x) - \mu_{1|0}(x)|] \\
 &\quad + \mathbb{E} [\mathbb{I}\{\hat{\delta}_u(x) \neq \delta_u(x)\} |\mu_{11|1}(x) - \mu_{1|0}(x)|] \\
 &= R_l(\hat{\delta}_l) + R_u(\hat{\delta}_u).
 \end{aligned}$$

Obviously $|\tilde{Q}(X_i) - \check{Q}(X_i)|$ is bounded between 0 and $|s|$, so by the Hoeffding inequation:

$$\Pr \left(\frac{1}{n} \sum_{i=1}^n |\tilde{Q}(X_i) - \check{Q}(X_i)| \leq |s| \cdot \left(R_l(\hat{\delta}_l) + R_u(\hat{\delta}_u) + \frac{t}{2\sqrt{n}} \right) \right) \geq 1 - \exp\left(-\frac{t^2}{2}\right).$$

Combining these two bounds via the union bound gives the result. $\square$

G. Additional Experimental Details and Results

G.1. Baseline: Uplift Modeling

Uplift modeling estimates the conditional average treatment effect (CATE) for individuals with specific features and is widely adopted in economics, precision medicine, decision making, and advertising placement. Many methods have been proposed for estimating CATE, such as outcome regression (OR), inverse propensity scoring (IPS), and doubly robust (DR) methods. The Uplift model is characterized by its exclusive focus on the single outcome Y , which is particularly suitable for our incentive allocation scenario where only the binary purchase behavior needs to be tracked. The detail specific estimators of Uplift model are:

$$\begin{aligned}\hat{\mu}_{1|1}^{OR}(X) &= \hat{\mu}_{1|1}(X), \hat{\mu}_{1|1}^{IPS}(X) = \frac{TY}{\hat{e}(X)}, \hat{\mu}_{1|1}^{DR}(X) = \hat{\mu}_{1|1}(X) + \frac{TY - \hat{\mu}_{1|1}(X)}{\hat{e}(X)}, \\ \hat{\mu}_{1|0}^{OR}(X) &= \hat{\mu}_{1|0}(X), \hat{\mu}_{1|0}^{IPS}(X) = \frac{(1-T)Y}{\hat{e}(X)}, \hat{\mu}_{1|0}^{DR}(X) = \hat{\mu}_{1|0}(X) + \frac{(1-T)Y - \hat{\mu}_{1|0}(X)}{\hat{e}(X)},\end{aligned}$$

where $\hat{\mu}_{1|1}(X)$ and $\hat{\mu}_{1|0}(X)$ are estimators obtained through regression. Uplift model issues incentives to the individuals who have high CATE, i.e., $\mu_{1|1}(X) - \mu_{1|0}(X)$.

G.2. Datasets

KuaiRec is a dataset from a short video platform, containing user-video information with watching ratio as the outcome of interest. In KuaiRec, we binarize the observed watching ratios by setting $Y = 1$ for ratios ≥ 0.6 and $Y = 0$ otherwise. Since KuaiRec is a fully exposed dataset, we randomly assign 40% of the observations to have $I(1) = 1$. The other operations and experimental details are consistent with those in Section 6.

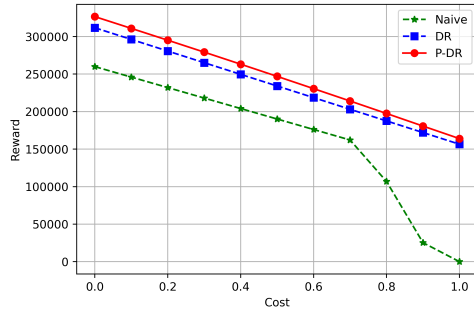
G.3. Additional results

The experimental results on KuaiRec are shown in Table 3 and Figure 4.

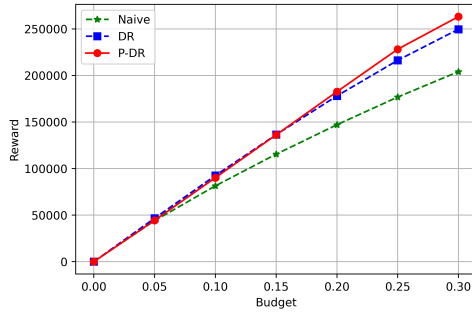
We further conduct experiments in other scenarios on datasets ML-1M and KuaiRec. Table 4 and Figures 5 and 6 show the results with high return of “No Claim - Incentive Buyer”, while Table 5 and Figures 7 and 8 present the results with “cash incentive” scenario that the intervention incurs a cost regardless of whether an individual accepts it or not.. Table 6 shows the results with mild violations of the monotonicity assumption (include a proportion of Y_{0010} and Y_{0110} individuals). Then we adjust the proportion of the l_2 stratum to $0.8\times$ and $1.2\times$ its original level. The l_1 stratum is proportionally reduced to preserve overall population balance, and all other settings remain unchanged. The results are shown in the Table 7. We also introduce an alternative data generation configuration, specifically by assigning individuals with higher predicted ratings as “Always Takers” to better reflect natural user affinity. The results are shown in the Table 8. Table 9 shows the results with the Neural Collaborative Filtering (NCF) model replaced by the Neural Matrix Factorization (NMF) model. The results show similar trends as those presented in Section 6, demonstrating the effectiveness of our proposed learning framework. Furthermore, Table 10 presents the results under completely unconstrained interventions. Finally, we benchmark our methods against the methods developed under the exclusion restriction assumption (Li et al., 2023b), with the comparison results reported in Table 11. These additional experiments consistently show that our method maintains its superiority across various complex scenarios.

Table 3. Performance comparison of naive, uplift modeling-based methods, and our proposed learning methods on KuaiRec, with $r_1, r_2, r_3, r_4, r_5, r_6$ set as $[0, 1.2, 0, 0, 0.6, -0.6]$ and cash as incentive.

COUPON		KuaiREC						
METHODS	Y ₀₀₀₀	Y ₀₀₀₁	Y ₀₀₁₁	Y ₀₁₀₀	Y ₀₁₀₁	Y ₀₁₁₁	RETURN	RI
NAIVE	20,904	109,354	21,535	28,900	130,522	9,122	204,065	–
OR	20,923	130,591	304	13,251	154,854	414	249,373	–
P-OR	2,151	135,184	14,422	18,737	142,379	7,464	243,170	-2.5%
IPS	30,343	121,050	362	16,550	151,453	579	235,784	–
P-IPS	9,278	142,072	385	11,573	156,688	341	264,295	12.1%
DR	21,630	130,045	93	13,519	154,980	70	249,000	–
P-DR	8,793	141,866	1,101	12,346	155,920	311	263,605	5.9%



(a) Effects of varying cost.



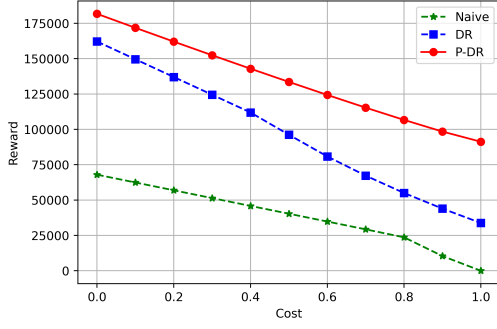
(b) Effects of varying budget.

Figure 4. Effects of varying cost and budget (KuaiRec).

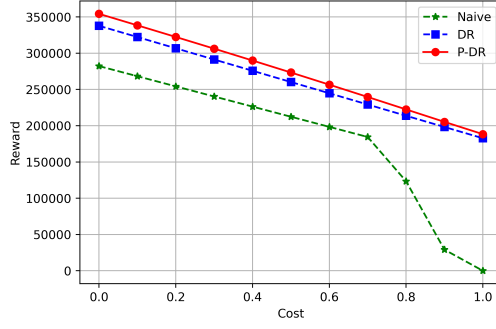
Table 4. Performance comparison of naive, uplift modeling-based methods, and our proposed learning methods on ML-1M and KuaiRec datasets, with $r_1, r_2, r_3, r_4, r_5, r_6$ set as $[0, 1.4, 0, 0, 0.6, -0.6]$ and coupon as incentive.

COUPON			ML-1M					
METHODS	Y_{0000}	Y_{0001}	Y_{0011}	Y_{0100}	Y_{0101}	Y_{0111}	RETURN	RI
NAIVE	31,785	17,222	236,563	50,931	46,013	9,341	22,297	–
OR	141,626	42,201	17,582	62,836	114,529	13,081	86,376	–
P-OR	40,018	67,330	111,022	52,040	107,601	13,844	116,234	34.6%
IPS	164,920	41,969	10,561	51,533	113,317	9,555	92,007	–
P-IPS	111,017	66,220	1,122	35,209	116,184	3,448	133,022	44.6%
DR	151,877	37,393	16,344	60,619	112,276	13,346	79,982	–
P-DR	111,816	68,687	3,231	37,511	117,629	4,008	135,593	69.6%

COUPON			KuaiREC					
METHODS	Y_{0000}	Y_{0001}	Y_{0011}	Y_{0100}	Y_{0101}	Y_{0111}	RETURN	RI
NAIVE	20,948	109,607	21,440	28,704	130,557	9,081	226,335	–
OR	20,583	130,909	301	13,247	154,889	408	275,961	–
P-OR	2,457	133,728	15,676	18,894	140,642	8,940	266,240	–3.5%
IPS	29,920	121,567	194	16,487	151,753	416	260,996	–
P-IPS	9,911	141,196	636	11,638	156,483	473	291,280	11.6%
DR	21,055	130,590	119	13,310	155,228	35	275,942	–
P-DR	9,373	141,022	1324	12,456	155,721	441	290,599	5.3%

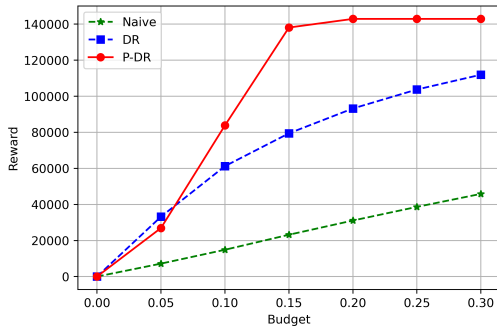


(a) Value Change on ML-1M.

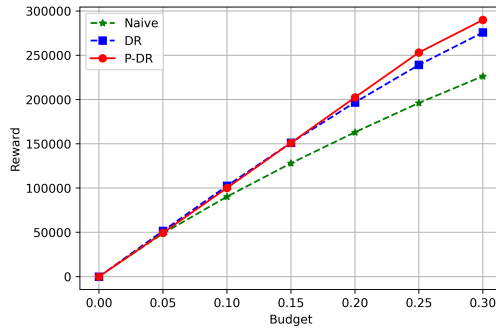


(b) Value change on KuaiRec.

Figure 5. Effects of varying cost (high return).



(a) Value change on ML-1M.



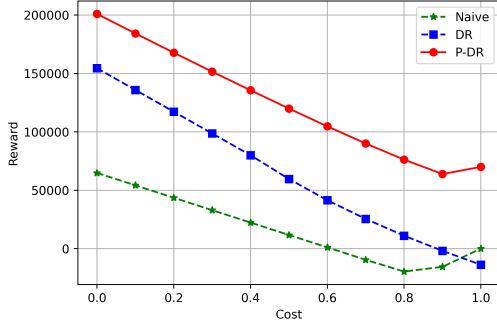
(b) Value change on KuaiRec.

Figure 6. Effects of varying budget (high return).

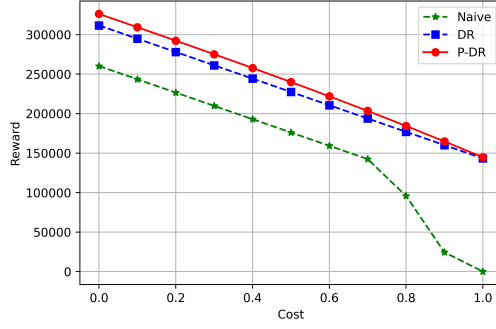
Table 5. Performance comparison of naive, uplift modeling-based methods, and our proposed learning methods on ML-1M and KuaiRec datasets, with $r_1, r_2, r_3, r_4, r_5, r_6$ set as $[0, 1.2, 0, -0.4, 0.6, -0.6]$ and cash as incentive.

CASH			ML-1M					
METHODS	Y_{0000}	Y_{0001}	Y_{0011}	Y_{0100}	Y_{0101}	Y_{0111}	RETURN	RI
NAIVE	31,785	17,222	236,563	50,931	46,013	9,341	22,297	-
OR	141,626	42,201	17,582	62,836	114,529	13,081	86,376	-
P-OR	40,018	67,330	111,022	52,040	107,601	13,844	116,234	34.6%
IPS	164,920	41,969	10,561	51,533	113,317	9,555	92,007	-
P-IPS	11,1017	66,220	1122	35,209	116,184	3,448	133,022	44.6%
DR	151,877	37,393	16,344	60,619	112,276	13,346	79,982	-
P-DR	111,816	68,687	3,231	37,511	117,629	4,008	135,593	69.5%

CASH			KUIREC					
METHODS	Y_{0000}	Y_{0001}	Y_{0011}	Y_{0100}	Y_{0101}	Y_{0111}	RETURN	RI
NAIVE	20,947	109,605	21,441	28,704	130,559	9,081	192,931	-
OR	20,585	130,910	301	13,249	154,883	409	244,477	-
P-OR	2,369	136,736	12,768	17,736	143,987	6,741	239,336	-2.1%
IPS	29,875	121,771	173	16,259	151,887	372	230,531	-
P-IPS	10,104	141,165	392	11,679	156,600	397	258,448	12.1%
DR	21,198	130,558	172	13,355	155,001	53	244,296	-
P-DR	9,716	141,063	888	12,270	156,031	369	257,765	5.5%

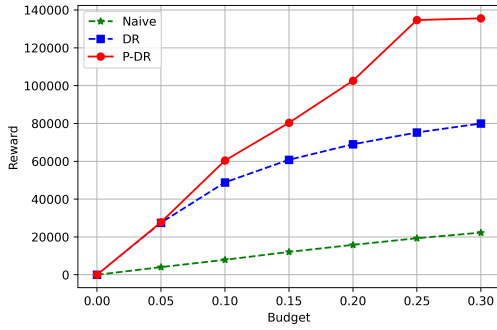


(a) Value change on ML-1M.

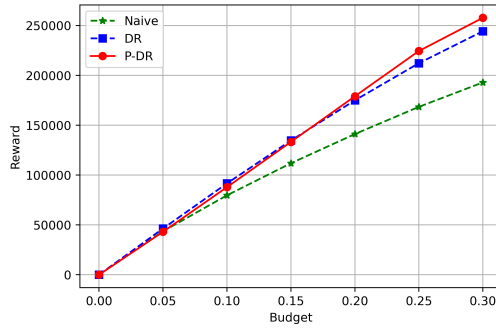


(b) Value change on KuaiRec.

Figure 7. Effects of varying cost (cash).



(a) Value change on ML-1M.



(b) Value change on KuaiRec.

Figure 8. Effects of varying budget (cash).

Table 6. Performance comparison of naive, uplift modeling-based methods, and our proposed learning methods on ML-1M, with mild violations of the monotonicity assumption.

COUPON		ML-1M							
METHODS	Y_{0000}	Y_{0001}	Y_{0011}	Y_{0100}	Y_{0101}	Y_{0111}	RETURN	RI	
NAIVE	40,803	20,450	196,434	44,886	54,940	5,585	54,153	–	
OR	151,133	44,593	11,652	56,759	116,102	9,898	117,234	–	
P-OR	41,591	67,327	108,279	51,843	109,020	13,350	138,194	17.9%	
IPS	173,527	44,983	5,208	45,462	115,813	6,302	119,686	–	
P-IPS	99,327	64,776	785	33,856	113,509	2,945	144,070	20.3%	
DR	162,434	40,944	8,857	53,465	115,860	9,112	113,182	–	
P-DR	100,492	68,762	3,358	37,957	115,459	3,661	149,593	32.2%	

Table 7. Performance comparison of naive, uplift modeling-based methods, and our proposed learning methods on ML-1M and KuaiRec datasets, with adjusting the proportion of the l_2 stratum to $0.8\times$ and $1.2\times$ its original level.

COUPON		ML-1M ($0.8\times$)						
METHODS	Y_{0000}	Y_{0001}	Y_{0011}	Y_{0100}	Y_{0101}	Y_{0111}	RETURN	RI
NAIVE	27,576	12,522	245,872	53,065	42,669	10,151	34,537	-
OR	148,561	30,842	19,515	64,622	114,426	13,889	97,333	-
P-OR	42,205	53,787	119,100	54,919	106,759	15,085	119,549	22.8%
IPS	174,892	31,250	11,182	51,647	113,478	9,406	99,943	-
P-IPS	125,335	54,224	967	34,964	115,800	3,460	132,473	32.6%
DR	142,616	25,899	17,717	60,046	107,402	13,768	87,259	-
P-DR	125,665	55,014	3,808	39,408	117,124	4,520	133,579	53.1%
COUPON		ML-1M ($1.2\times$)						
METHODS	Y_{0000}	Y_{0001}	Y_{0011}	Y_{0100}	Y_{0101}	Y_{0111}	RETURN	RI
NAIVE	34,314	22,614	228,238	49,216	48,734	8,739	51,134	-
OR	135,255	53,551	15,547	60,854	114,251	12,397	125,374	-
P-OR	40,235	81,224	95,228	52,378	109,330	13,460	154,991	23.6%
IPS	154,296	53,919	9,568	50,633	114,223	9,216	127,707	-
P-IPS	115,988	76,940	766	33,633	114,405	3,114	159,103	24.6%
DR	143,526	49,294	14,798	58,858	112,652	12,727	119,108	-
P-DR	117,589	82,521	3,904	38,783	116,416	4,470	166,193	39.5%

Table 8. Performance comparison of naive, uplift modeling-based methods, and our proposed learning methods on Jester, with assigning individuals with higher predicted ratings as "Always Takers".

COUPON		JESTER						
METHODS	Y_{0000}	Y_{0001}	Y_{0011}	Y_{0100}	Y_{0101}	Y_{0111}	RETURN	RI
NAIVE	2	15,576	20,378	40,148	83,615	51,772	37,797	-
OR	11	24,511	773	65,715	110,783	9,698	90,064	-
P-OR	8,159	105,287	10,618	52,235	34,912	280	147,124	63.4%
IPS	279	19,359	3,928	63,508	100,554	23,863	69,245	-
P-IPS	591	45,869	14,150	46,493	102,126	2,262	114,961	66.0%
DR	158	19,819	945	69,192	111,196	10,181	84,392	-
P-DR	590	46,020	13,608	46,945	102,442	1,886	115,558	36.9%

Table 9. Performance comparison of naive, uplift modeling-based methods, and our proposed learning methods on ML-1M, with the Neural Collaborative Filtering (NCF) model replaced by the Neural Matrix Factorization (NMF) model.

COUPON		ML-1M						
METHODS	Y_{0000}	Y_{0001}	Y_{0011}	Y_{0100}	Y_{0101}	Y_{0111}	RETURN	RI
NAIVE	27,059	18,400	218,846	53,575	63,622	10,353	54,041	-
OR	54,312	53,517	121,361	47,295	105,077	10,293	121,091	-
P-OR	30,610	66,525	126,995	58,579	96,836	12,290	130,558	7.8%
IPS	163,562	42,209	10,387	52,279	113,626	9,772	112,963	-
P-IPS	90,835	65,535	998	34,419	113,556	3,528	144,659	28.1%
DR	153,079	38,384	15,486	60,278	111,524	13,084	105,125	-
P-DR	58,328	46,490	3,804	36,486	108,388	4,191	118,306	12.5%

Table 10. Performance comparison of naive, uplift modeling-based methods, and our proposed learning methods on ML-1M under the scenario of completely unconstrained interventions.

COUPON		ML-1M					RETURN	RI
METHODS	Y_{0000}	Y_{0001}	Y_{0011}	Y_{0100}	Y_{0101}	Y_{0111}		
NAIVE	165,374	39,400	454,508	114,548	109,667	32,375	93,655	–
OR	192,728	46,826	21,447	69,499	124,047	14,847	121,711	–
P-OR	78,364	68,826	410,586	79,775	118,715	21,678	140,813	15.7%
IPS	213,223	45,366	13,410	56,852	121,187	10,897	120,613	–
P-IPS	114,361	66,331	1,130	35,320	116,422	3,449	147,381	22.2%
DR	157,789	37,695	16,776	61,409	113,339	13,548	105,109	–
P-DR	115,130	68,697	3,672	39,840	117,821	4,479	150,442	43.1%

Table 11. Performance comparison of naive, uplift modeling-based methods, methods developed under the exclusion restriction assumption (denoted with the prefix “Li-”) and our proposed learning methods on Jester.

COUPON		JESTER					RETURN
METHODS	Y_{0000}	Y_{0001}	Y_{0011}	Y_{0100}	Y_{0101}	Y_{0111}	
NAIVE	10,811	13,822	2,423	77,285	100,844	6,305	77,310
OR	42	20,148	9	47,119	137,759	6,414	102,985
LI-OR	7,461	105,959	31,487	38,109	26,880	1,595	142,322
P-OR	26,550	106,534	10,450	22,924	39,252	5,781	147,923
IPS	112	13,009	3	57,555	127,026	13,786	83,555
LI-IPS	520	44,842	21,226	45,035	97,541	2,327	110,939
P-IPS	2,770	74,942	197	41,605	80,456	11,521	131,291
DR	138	18,130	15	48,675	137,400	7,133	99,916
LI-DR	517	44,882	20,816	45,254	97,965	2,057	111,403
P-DR	2,781	75,080	198	41,475	80,453	11,504	131,465